

SPECIFICATIONS FOR EXPRESS SCHEDULE FILE FORMAT

This manual applies to the following
JEOL products:

JBX-5000LS,	JBX-5000LS/E
JBX-6000FS,	JBX-6000FS/E
JBX-9000MV,	JBX-9000MV/II
JBX-9300FS,	JBX-3030MV
JBX-3040MV,	JBX-3050MV
JBX-6300 series	JBX-5500 series

SPECIFICATIONS FOR EXPRESS JSCHEDULE FILE FORMAT

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or maintenance for the proper use of the instrument.**

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NOTATIONAL CONVENTIONS AND GLOSSARY

■ Examples for general notations

– CAUTION – :	Important precautions for use, which, if not followed, may result in damage to or problems with the device itself.
 :	Additional points to remember regarding the operation.
 :	A reference to another section, chapter or manual.
1, 2, 3 :	Numbers indicate a series of operations that achieve a task.
 :	A diamond indicates a single operation that achieves a task.
File :	The names of menus, commands, or parameters displayed on the screen are denoted with bold letters.
File–Exit :	Selecting a menu item from a pulldown menu is denoted by linking the menu and the item with a dash (–). For example, File–Exit means selecting Exit from the File menu.
 :	Keys on the keyboard are denoted by enclosing their names in a box.

■ Examples for mouse terminology

Mouse pointer:	A mark, displayed on the screen, which moves following the movement of the mouse. It is used to specify a menu item, command, parameter value, and other items. Its shape changes according to the situation.
Click:	To press and release the left mouse button.
Right-click:	To press and release the right mouse button.
Double-click:	To press and release the left mouse button twice quickly.
Drag:	To hold down the left mouse button while moving the mouse.

CONTENTS

SAFETY PRECAUTIONS

1	GENERAL	1
2	COMPOSITION	2
2.1	File Format	2
2.2	Command Format	3
3	OUTLINE OF COMMANDS AND CONTROL CHARACTERS	5
3.1	Outline of Commands	5
3.2	Outline of Control Characters	8
4	FORMATS AND FUNCTIONS OF COMMANDS AND CONTROL CHARACTERS.....	9
4.1	Format of Schedule Command	9
4.2	Header Commands	10
4.2.1	MAGAZIN	10
4.2.2	MTRL.....	10
4.3	Cassette Block Commands	12
4.3.1	#.....	12
4.3.2	%	13
4.3.3	LYR	14
4.3.4	JDF	14
4.3.5	EOS	15
4.3.6	ACC.....	16
4.3.7	CALPRM	16
4.3.8	DEFMODE	17
4.3.9	RESTOR	17
4.3.10	OFFSET	19
4.3.11	MCOR.....	20
4.3.12	HEIGHT	22
4.3.13	SUBHEI	23
4.3.14	SHTCOR	24
4.3.15	STEPPER	25
4.3.16	MMDPRM	25
4.3.17	SHTOFS	26
4.3.18	SPCPRM	27
4.3.19	PECPRM	27
4.3.20	FOGPRM	28
4.3.21	PRCPRM	29
4.3.22	PRJPRM	29
4.3.23	LECPRM	30
4.3.24	GLECPRM	30
4.3.25	ANG2SIZ	31

4.3.26	SSCPRM	31
4.3.27	PEDPRM	32
4.3.28	WARMUP	33
4.3.29	LBC	34
4.3.30	HSWITCH	35
4.3.31	GLMPRT	36
4.3.32	PLMPRT	37
4.3.33	CHMPRT	38
4.3.34	GLMDET	40
4.3.35	PLMDET	41
4.3.36	CHMDET	42
4.3.37	WFR3M	43
4.3.38	CHIPAL	44
4.3.39	CDRIFT	45
4.3.40	PDRIFT	46
4.3.41	TDRIFT	47
4.3.42	PATH	48
4.3.43	P	49
4.3.44	SPPRM	51
4.3.45	STGPRM	53
4.3.46	C	55
4.3.47	RESIST	55
4.3.48	SHOT	56
4.3.49	MODULAT	58
4.3.50	MMDOFS	60
4.3.51	DEFOCUS	63
4.4	FOOTER COMMAND	64
4.4.1	END	64
4.5	CONTROL CHARACTERS	65
4.5.1	– (hyphen)	65
4.5.2	; (semicolon)	65
4.5.3	@ (at mark)	66

5 APPENDICES 67

5.1	Limits of Commands	67
5.2	Limits of Parameters	68
5.2.1	For JBX-6000FS	68
5.2.2	For JBX-9000MV	75
5.2.3	For JBX-9300FS	82
5.2.4	For JBX-3030MV	89
5.2.5	For JBX-6000FS/E	97
5.2.6	For JBX-6300 series	104
5.2.7	For JBX-5500 series	111
5.3	Writable Characters	118
5.4	Examples of Schedules	119

SAFETY PRECAUTIONS

Before using the instrument, be sure to read this instruction manual, and upon fully understanding its contents, use the instrument properly. This section, “Safety Precautions”, contains important information about safety.

- **The safety indications and their meanings are as follows:**

	DANGER:	An imminently hazardous situation which, if not avoided, will result in death or serious injury.
	WARNING:	A potentially hazardous situation which, if not avoided, could result in death or serious injury.
	CAUTION:	A potentially hazardous situation which, if not avoided, may result in minor or moderate injury, or a situation that could result in serious damage to facilities or acquired data.

- **Labels bearing the following symbols are attached to dangerous locations on the instrument. Do not touch any of these locations with your hands or anything else.**



Examples of symbols

- Use the instrument properly within the scope of the purpose and usage described in its brochures and manuals.
- Never remove or disable protective parts, safety devices, or other safety features.
- Never make modifications that include disassembling the instrument or installing substitute parts.
- Never disconnect the grounding wire or move it from the prescribed position. Failure to follow this instruction could result in electric shock.
- When you dispose of the instrument or liquid or other waste, follow all applicable laws and regulations, and dispose of it in a proper manner without polluting the environment.
- Be sure to read the “Safety Precautions” section of the manuals for the accessories attached to or built into the instrument.
- If anything is unclear, please contact your local JEOL service office.

1 GENERAL

This manual specifies the formats of the writing schedule files of the EXPRESS (this stands for exposure preparation system for scheduling), the commands, and the parameters used in a JEOL electron beam lithography system.*

A schedule in the electron beam lithography system is a series of commands and parameters that include jobdeck files, specification of the number of the cassette, arrangement of chip patterns and character strings to be written on a substrate, writing conditions, writing order, and pattern writing. The schedule is treated as a schedule file.

Since, in actual practice, schedule files are to be created by the user, this manual deals with commands and parameters that relate to assignments of jobdecks to substrates, corrections of hardware conditions, and control of the hardware.

A schedule file consisting of a header, cassette blocks, and a footer is to be created for every magazine corresponding to one unit of pattern writing. The parameters that do not vary depending on the cassette in the magazine are written in the header and footer. The parameters that are appropriate to individual cassettes are written in those cassette blocks. For instance, the magazine name is specified in the header, the parameters related to writing control are specified in the cassette blocks, and the instruction to end the e-beam writing is given in the footer.

The schedule files can be created and edited by using the file editor utilities of various computer systems. Schedule commands are expressed in ASCII using numerals, uppercase letters, and several special symbols.

* JEOL electron-beam lithography systems are of various types. Unless otherwise noted, the model name JBX-6000FS written in the text or footnotes also applies to models such as the JBX-5DII, JBX-5FE, and JBX-5000LS. Similarly, the JBX-6000FS/E also applies to models such as the JBX-5000LS/E, the JBX-9000MV also applies to the JBX-9000MV/II, and the JBX-3030MV also applies to the JBX-3040MV and the JBX-3050MV, the JBX-6300series also applies to the JBX-6300xx (xx: Last characters following JBX-6300.), and the JBX-5500series also applies to the JBX-5500xx (xx: Last characters following JBX-5500.) Electron Beam Lithography System.

2 COMPOSITION

2.1 File Format

A schedule file is created on a magnetic recording medium, such as a disk or a magnetic tape, and has the following format:

- Sequential
- Formatted
- Record size: Variable length (up to 255 Bytes/record)
- Record attribute: Carriage return/carriage control

The name of a schedule file is limited as follows:

- File name: The name that the user assigns to the magazine; a character string that consists of any letters, numbers, and/or symbols
- Number of characters: Not limited
- Extension: “.SDF” or “.sdf”

Example: abcd_01.sdf

The filename can be any character string consisting of alphanumeric characters and symbols allowed by the computer system. The filename extension (called “file type” in this manual) can be either “.SDF” or “.sdf.”, for example, (magazine_name).SDF or (magazine_name).sdf.

A schedule file is to be prepared corresponding to each magazine to be written.

As mentioned in Chapter 1, a schedule file consists of one header, one footer, and one or more cassette blocks. The parameters that are not related to each cassette in the magazine are written in the header and footer. The parameters inherent to individual layers are written in the cassette blocks.

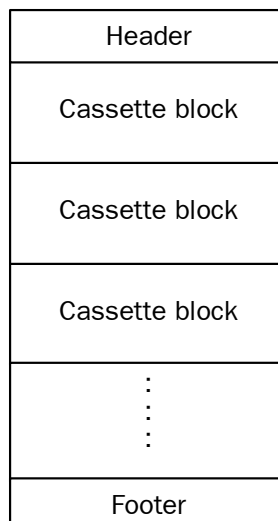


Fig 1 Schedule file format

One record in a schedule file is equivalent to one physical line in which a schedule command parameter line is stored. A schedule command parameter line can be divided into two or more contiguous physical lines, which form a logical line. A schedule file consists of multiple logical lines.

2.2 Command Format

Various parameters in a schedule are specified by using the appropriate commands. Schedule commands are hierarchically constructed as shown below.

Table 1 Definition of Symbols

Symbol	Description
=	Item on the left side is composed of elements on the right side.
+	Indicates sequence of commands.
[]	Elements in square brackets [] can be used zero or more times.
【 】	Element in thick brackets 【 】 can be used zero or one time. Although the same element can be designated two or more times, only the last one is in effect.
[]	Elements in thin brackets [] can be used one or more times.
{ }	One of the elements separated by vertical bars within braces { } must be selected.
{ / }	Elements separated by slashes / within braces { } can be used in an arbitrary order.
< >	Element within angle brackets < > represents other commands.

Schedule file = <Header> + [<Cassette block>] + <Footer>

<Header> = MAGAZIN + 【MTRL】

<Cassette block> = # + 【%】 + {LYR | JDF}
+ { 【EOS】 | 【ACC*】 + 【CALPRM】 } + 【DEFMODE】 + 【RESTOR】
+ { [<Writing correction condition definition>]
/ [<Writing control parameter definition>]
/ [<Jobdeck parameter definition>] }

<Writing correction condition definition> = { 【OFFSET】 / 【MCOR】
/ 【HEIGHT】 / 【SUBHEI】 / 【SHTCOR】
/ 【STEPPER】 / 【MMDPRM】
/ 【SHTOFS】 / 【SPCPRM】 / 【PECPRM】
/ 【FOGPRM】 / 【PRCPRM】 / 【PRJPRM】
/ 【LECPRM】 / 【GLECPRM】 / 【ANG2SIZ】 /
【SSCPRM】 / 【PEDPRM】 }

<Writing control parameter definition> = { 【WARMUP】 / 【LBC】 / 【HSWITCH】
/ [<Mark protection definition>]
/ [<Mark detection control>]
/ { 【WFR3M】 | 【CHIPAL】 } }

* ACC command is indispensable in JBX-9300FS.

/【<Drift control>】}

$$\langle \text{Mark protection definition} \rangle = \{ \text{[GLMPRT]} / \text{[PLMPRT]} / \text{[CHMPRT]} \}$$
$$\langle \text{Mark detection control} \rangle = \{ [\text{GLMDET}] / [\text{PLMDET}] / [\text{CHMDET}] \}$$
$$\langle \text{Drift control} \rangle = \{ \text{CDRIFT} / \text{PDRIFT} / \text{TDRIFT} \}$$
$$\begin{aligned} \text{<Jobdeck parameter definition>} &= \{ \text{【PATH】} / [\text{<Writing pattern definition>}] \\ &\quad / [\text{<Writing character string definition>}] \\ &\quad / [\text{<Writing condition definition>}] \} \end{aligned}$$
$$\langle \text{Writing pattern definition} \rangle = P + \{ \text{【SPPRM】} / \text{【STGPRM】} \}$$

<Writing character string definition> = C

$$\langle \text{Writing condition definition} \rangle = \{ [\text{RESIST}] / [\text{SHOT}] / [\text{MODULAT}] / [\text{MMDOFS}] / [\text{DEFOCUS}] \}$$

<Footer> = END

3 OUTLINE OF COMMANDS AND CONTROL CHARACTERS

3.1 Outline of Commands

The schedule commands are shown in the tables below.

Table 2 Schedule commands (header)

No.	Command	Required/Optional	Description
(1)	MAGAZIN	Required	Specifies both the beginning of the schedule and the ID name of the magazine (magazine name) that is to be a unit of pattern writing.
(2)	MTRL	Optional	Specifies the supply source or method of supplying of substrates.

Table 3 Schedule commands (cassette block)

No.	Command	Required/Optional	Description
(1)	#	Required	Specifies both the beginning of the cassette block and the shelf numbers of substrates to be written.
(2)	%	Optional	Specifies the substrate position on the multi holder cassette.
(3)	LYR	Either this or JDF required	Specifies the filename of the compiled jobdeck file.
(4)	JDF	Either this or LYR required	Specifies both the filename and layer number of the jobdeck file that is not yet compiled.
(5)	EOS	Optional	Specifies both the EOS mode and table number to be used in e-beam writing. (To be used exclusively for JBX-6000FS, JBX-6000FS/E, JBX-6300 series, JBX-5500 series)
(6)	ACC	Required	Specifies the accelerating voltage used in e-beam writing. (To be used exclusively for JBX-9300FS.)
(7)	CALPRM	Optional	Specifies the name of the hardware calibration condition to be referred to in e-beam writing. (To be used exclusively for JBX-6000FS, JBX-6300 series and JBX-5500 series.)
(8)	DEFMODE	Optional	Specifies the number of deflection stages for e-beam writing. (To be used exclusively for JBX-9300FS.)
(9)	RESTOR	Optional	Specifies setting the electron optical system (EOS) to the prestored lens values. (To be used exclusively for JBX-9300FS, JBX-6000FS, JBX-6300 series and JBX-5500 series.)
(10)	OFFSET	Optional	Specifies the amount of shift of the origin in the substrate-center coordinate system.
(11)	MCOR	Optional	Specifies the amount of orthogonal correction for the substrate stage.
(12)	HEIGHT	Optional	Specifies the amount of height correction for the substrate using the main deflector.
(13)	SUBHEI	Optional	Specifies the amount of height correction for the substrate

No.	Command	Required/Optional	Description
			using the sub deflector.
(14)	SHTCOR	Optional	Specifies the amount of shot time correction for each shape in the patterns to be written.
(15)	STEPPER	Optional	Specifies the name of a file in which stepper distortion correction amounts are stored. (To be used exclusively for JBX-9000MV.)
(16)	MMDPRM	Required	Specifies execution of the mix-and-match distortion correction function in e-beam writing. (To be used exclusively for JBX-9300FS.)
(17)	SHTOFS	Optional	Specifies the name of a file in which the correction amounts are stored for dose amount changes for individual beam sizes.
(18)	SPCPRM	Optional	This command specifies the shot-position correction writing for each shot size, and specifies the name of the file that contains the correction parameter settings to use for that writing. (Available in JBX-3030MV and later models.)
(19)	PECPRM	Optional	Specifies the name of a file in which various parameters for proximity effect correction writing are stored. (Available in JBX-9000MV and JBX-3030MV and later models.)
(20)	FOGPRM	Optional	Specifies the name of a file in which various parameters for fogging error correction writing are stored. (Available in JBX-9000MV and JBX-3030MV and later models.)
(21)	PRCPRM	Optional	Specifies the name of a file in which various parameters for process error correction writing are stored. (Available in JBX-9000MV and JBX-3030MV and later models.)
(22)	PRJPRM	Optional	Specifies the name of a file in which various parameters for projection error correction writing are stored. (Available in JBX-9000MV and JBX-3030MV and later models.)
(23)	LECPRM	Optional	This command specifies the Microloading-effect correction writing and specifies the name of the file that contains the correction parameter settings to use for that writing. (Available in JBX-3030MV and later models.)
(24)	GLECPRM	Optional	This command specifies the Global loading effect correction writing and specifies the name of the file that contains the correction parameter settings to use for that writing. (Available in JBX-3040MV and later models.)
(25)	ANG2SIZ	Optional	Executes trapezoid correction writing at arbitrary angles and specifies the name of a file in which various parameters for trapezoid correction writing are stored. (Available in JBX-3030MV and later models.)
(26)	SSCPRM	Optional	This command specifies the shot-size correction writing for each position in the field plane, and specifies the name of the file that contains the correction parameter settings to use for that writing. (Available in JBX-3030MV and later models.)

No.	Command	Required/Optional	Description
(27)	PEDPRM	Optional	Specifies the time-dependent dose correction writing, and the name of the file containing the correction parameters that apply to the writing. (Available in JBX-3050MV and later models.)
(28)	WARMUP	Optional	Specifies the waiting time before starting the e-beam writing after loading a substrate in the workstage.
(29)	LBC	Optional	Specifies the operation mode of the LBC (laser beam control).
(30)	HSWITCH	Optional for direct writing	Specifies the operation mode for substrate height measurement used in direct writing.
(31)	GLMPRT	Optional for direct writing and PSM writing*	Specifies the parameters for global mark protection.
(32)	PLMPRT	Optional for PSM writing	Specifies the parameters for plate mark protection.
(33)	CHMPRT	Optional for direct writing	Specifies the parameters for chip mark protection.
(34)	GLMDET	Optional for direct writing and PSM writing	Specifies the operation mode for the writing control program, which is applied to global mark detection.
(35)	PLMDET	Optional for PSM writing	Specifies the operation mode for the writing control program, which is applied to plate mark detection.
(36)	CHMDET	Optional for direct writing	Specifies the operation mode for the writing control program, which is applied to chip mark detection.
(37)	WFR3M	Optional for direct writing	Specifies the mode for chip mark detection and writing correction. (To be used exclusively for JBX-6000FS, JBX-6300 series and JBX-5500 series.)
(38)	CHIPAL	Optional for direct writing	Specifies the mode for chip mark detection and writing correction. (Invalid except for JBX-6000FS and JBX-5500 series.)
(39)	CDRIFT	Optional	Specifies the tolerance for beam current change during pattern writing.
(40)	PDRIFT	Optional	Specifies the tolerance for beam position change during pattern writing.
(41)	TDRIFT	Optional	Specifies the tolerance for substrate temperature change during pattern writing.
(42)	PATH	Optional	Specifies the menu name of the CALIB that is to be carried out for every PATH specified in the jobdeck.
(43)	P	Optional	Specifies the pattern data to be written at the array point to which the pseudo pattern number specified in the jobdeck is assigned, and also specifies the special writing mode for each writing pattern.
(44)	SPPRM	Optional	Specifies the parameters for shot partition in each writing pattern.

* PSM writing stands for phase shift mask writing.

No.	Command	Required/Optional	Description
(45)	STGPRM	Optional	Specifies the method for moving the substrate stage during pattern writing. (Available in JBX-3050MV and later models.)
(46)	C	Optional	Specifies the contents of the writing character string that corresponds to the character string number specified in the jobdeck.
(47)	RESIST	Optional	Specifies the sensitivity of the resist to be applied to pattern writing.
(48)	SHOT	Optional	Specifies the amount of writing time per shot of the electron beam.
(49)	MODULAT	Optional	Specifies the contents of the shot modulation table written in the jobdeck.
(50)	MMDOFS	Required	Specifies the position whose correction amount is to be applied in the mix-and-match distortion correction map to writing chips and various marks. (To be used exclusively for JBX-9300FS.)
(51)	DEFOCUS	Optional	Specifies an increment or decrement value for adjusting the focus lens.

Table 4 Schedule commands (footer)

No.	Command	Required/Optional	Description
(1)	END	Required	Specifies the end of the schedule. Also specifies the shelf number of the substrate to be loaded on the stage after pattern writing is completed.

3.2 Outline of Control Characters

The control characters that can be used in schedule files are shown in the table below.

Table 5 Control characters

No.	Control character	Required/Optional	Description
(1)	–	Optional	Specifies that this line is a continuation from the previous command parameter line.
(2)	;	Optional	Specifies that the character string that follows this semicolon in the line (record) is a comment.
(3)	@	Optional	Specifies a position where a file is to be inserted.

4 FORMATS AND FUNCTIONS OF COMMANDS AND CONTROL CHARACTERS

4.1 Format of Schedule Command

A typical format for a schedule command is shown below:

Typical format of a schedule command

【Table:△】		Command	△	Parameter List
①	②	③	②	④

① Table:

When the MODULAT command is to be specified, enter here a name with a colon (:) that identifies the MODULAT command.

☞ See the description of the MODULAT command in Sect. 4.3.46.

② △

Delimiter that consists of one or more consecutive spaces or tabs; it separates the table from the command, and the command from the parameter list.

③ Command

Enter here one of the commands listed in Chapter 3: “Outline of Commands and Control Characters.”

④ Parameter list

Enter here the parameter(s) related to the command given. If a parameter list consists of multiple parameters, separate adjacent parameters with a comma (,).

Although one physical line can contain up to 255 characters, the system ignores the characters after the 132nd in one line. When you wish to write a command parameter line that exceeds 132 characters, put a hyphen (–) in the first column of the succeeding line to declare continuation of the command parameter line, insert a delimiter (△), and write the rest of the parameter list.

A carriage return code (CR/LF), or a comment control code (;) ends a command parameter line. However, if the command parameter line continues to the next line, the system will ignore it.

4.2 Header Commands

4.2.1 MAGAZIN

(Required)

Format:

MAGAZIN 'm'
①

Example:

MAGAZIN 'MAGNAME'

- **Descriptions of the examples**

The assigned magazine name is MAGNAME.

- **Description of the command**

The MAGNAME command specifies the beginning of the schedule, and the identifier (magazine name) of the magazine that is a unit of pattern writing. This command is always put at the beginning of a series of schedule command lines and must not be omitted.

① m

Specifies the identifier (magazine name) of the magazine that represents a unit of pattern writing.

4.2.2 MTRL

(Optional)

Format:

MTRL m
①

Examples:

MTRL	STKR	(1)
MTRL	CRR	(2)
MTRL	ILN	(3)
MTRL	ALD	(4)

- **Descriptions of the examples**

- (1) Specifies that substrates stored in the stocker of the substrate handling system* are to be used for pattern writing.
- (2) Specifies that substrates stored on the carrier of the substrate handling system are to be used for pattern writing.
- (3) Specifies that substrates that are continuously supplied by the in-line processing system to the substrate handling system are to be used for pattern writing.

* For details about the substrate handling system, see the separate instruction manual.

- (4) Specifies that substrates stored in the autoloader (ALD) of your e-beam lithography system are to be used for pattern writing.

● **Description of the command**

This command specifies the supply source or method of supplying of substrates. If this command is omitted, the substrates stored in or supplied to the system will be used for pattern writing according to the default specified by the system.

① **m**

Specifies the container in which the substrates are stored or the supply method.

m	Description
STKR	Uses substrates stored in the stocker of the substrate handling system.
CRR	Uses substrates stored on the carrier of the substrate handling system.
ILN	Uses substrates continuously supplied by in-line processing to the substrate handling system.
ALD	Uses substrates stored in the autoloader (ALD) of your e-beam lithography system.

☞ For information about the substrate-handling system, refer to “General Statement” and “Lithography Operations” (or “Operation”) for the lithography system.

4.3 Cassette Block Commands

4.3.1

(Required)

Format:

#n [-m] [, n' [-m'], ...]

① ②

Example:

#4-6,4-2

- **Description of the example**

Pattern writing is to be carried out in the following order of shelf numbers, $2 \rightarrow 3 \rightarrow 4 \rightarrow 4 \rightarrow 5 \rightarrow 6$, according to the schedule of the cassette block that follows this command.

- **Description of the command**

Specifies the beginning of a cassette block and the substrates to be used for pattern writing. What to be written is specified by the schedule of the cassette block that follows this command. In accordance with the MTRL command of the header, this command specifies substrates by assigning the shelf numbers of the container of the substrate handling system or the autoloader (ALD) in which the substrates are stored. The same schedule can be used for pattern writing of more than one substrate. To do this, specify more than one shelf number.

The writing of substrates is carried out in ascending order, that is, from the substrate having the smallest shelf number to the one having the largest shelf number.

① n

Specifies the shelf number of the stocker of the substrate handling system or the autoloader (ALD) of the e-beam lithography system in which the substrate to be written is stored.

Form	Meaning
n, n'	n = n', the pattern is to be written twice on substrate n.

② m

Specifies the end of a continuous range of shelf numbers, beginning with n, of the container of the substrate handling system or the autoloader (ALD) in which the substrates to be written are stored, when the same schedule is used for pattern writing of more than one substrate.

Form	Meaning
n-m	n = m, the pattern is to be written only once on substrate n.

- ✍ For the lithography system having no autoloader, the system ignores the specification of a shelf number n.
In the two continuously specified cassette blocks, when the maximum of shelf numbers n, n', and m in the former cassette block is equal to the minimum of shelf numbers n, n', and m in the latter cassette block, the last substrate used in the former cassette block will be used with the first pattern writing schedule of the latter cassette block without changing the substrate.
- ✍ When writing patterns on substrates is continuously supplied by the in-line processing (that is, when you assign ILN in the MTRL statement), pay attention to the following items.
 - (1) In the two continuously specified cassette blocks, you cannot specify the minimum number in the latter cassette block shelf number of n, n', or m if the number is smaller than the maximum number in the former cassette block shelf number of n, n', or m.
 - (2) In a # command line, the shelf number n' should be the same number for the shelf number n or one more than the preceding shelf number n or n'.
 - (3) The shelf number m may not be smaller than the shelf number n. In the processing, $m - n + 1$ will be the number of substrates to be written continuously in one cassette block.

4.3.2 %

(Optional: inapplicable in JBX-5500 series.)

Format:

% s p

①②

Example:

% 2A

● Description of the example

Pattern writing is to be carried out on the substrate at position 2A of the multi-holder cassette according to the schedule of the cassette block that follows this command.

● Description of the command

The % command specifies the position of the writing substrate in the multi-holder cassette using an identification character string that designates the substrate position in the multi-holder cassette.

① s

Specifies the first identification character that designates the substrate position.

② p

Specifies the last identification character that designates the substrate position.

- ✍ The coordinates of the substrate position are specified in the system parameter file.

4.3.3 LYR

(Either LYR or JDF is required.)

Format:	
LYR	'1' ①
Example:	
LYR	'general_01'

- **Description of the example**

The patterns in the compiled jobdeck file `general_01.lyr` are to be used for pattern writing.

- **Description of the command**

The LYR command specifies the layer of the jobdeck to be written by assigning the name of the compiled jobdeck file (without the file type “.lyr”).

① l
Specifies a compiled jobdeck filename.

 Either the LYR command or the JDF command is required, but you may not specify both.

4.3.4 JDF

(Either LYR or JDF is required.)

Format:	
JDF	'j',n ① ②
Example:	
JDF	'general',1

- **Description of the example**

The patterns in the jobdeck specified by the layer No.1 of the jobdeck file `general.jdf` are used for pattern writing.

- **Description of the command**

This command specifies the layer of the jobdeck to be written by assigning the jobdeck filename (without the file type “.jdf”) and the layer number.

① j
Specifies the jobdeck filename.

② n
Specifies a layer number

 Either the LYR command or the JDF command is required but you may not specify both.

4.3.5 EOS

(Optional; applicable only in JBX-6000FS, JBX-6000FS/E, JBX-6300 series and JBX-5500 series)

Format:			
EOS	m, {t 'c' }		
	① ② ③		
Example:			
EOS	2, 4		(1)
EOS	2, 'DEFAULT'		(2)

● Description of the example

- (1) E-beam writing is to be carried out using the writing conditions and the calibration (CALIB) conditions designated by the EOS mode number 2 and EOS table number 4.
- (2) E-beam writing is to be carried out using the writing conditions and the calibration conditions designated by the EOS mode number 2 and the calibration condition name DEFAULT.

● Description of the command

The EOS command specifies hardware correction parameters for e-beam writing and calibration conditions using a combination of the EOS mode and the EOS table or calibration condition name.

Using the CALIB program, various correction factors for the hardware of the e-beam lithography system are sorted according to their writing conditions and calibration conditions, and stored in the storage for the EOS modes and the EOS tables or calibration condition names, respectively.

If the EOS statements are given in both the layer block of the jobdeck file and the cassette block of the schedule file, the specification in the schedule file will override that in the jobdeck file.

① m

Specifies the EOS mode number.

② t

Specifies the EOS table number.

 In JBX-6000FS, specify the EOS table.

③ c

Specifies the calibration condition name.

 In JBX-6000FS/E, JBX-6300 series and JBX-5500 series, specify the calibration condition name.

Selects “t” or “c” depend on the JBX equipment.

 This EOS command must be given either in schedule cassette blocks or in jobdeck layer blocks.

4.3.6 ACC

(Applicable only in (and required in) JBX-9300FS)

Format:	
ACC	v ①
Example:	
ACC	50

- **Description of the example**

The accelerating voltage of 50 kV is specified for the e-beam writing.

- **Description of the command**

The ACC command specifies the accelerating voltage for the e-beam writing.

① v

Specifies the accelerating voltage (kV) for the e-beam writing.

4.3.7 CALPRM

(Optional; inapplicable in JBX-6000FS, JBX-6300 series and JBX-5500 series.)

Format:	
CALPRM	'c' ①
Example:	
CALPRM	'CCURRE9'

- **Description of the example**

Various correction parameters for e-beam writing are to be acquired from the calibration conditions file CCURRE9.

- **Description of the command**

The CALPRM command specifies the name of the file in which are stored the system calibration conditions for various hardware correction parameters such as conditions for e-beam writing and for calibration to be used for writing patterns. The names of the system calibration condition files are set using the calibration program. The system correction parameters for e-beam writing can be changed by specifying the name of the various calibration condition file to be used. When this command is omitted, the default filename registered in the system will be used.

① c

Specifies the various calibration condition filename.

4.3.8 DEFMODE

(Optional; applicable only in JBX-9300FS)

Format:

DEFMODE m
①

Example:

DEFMODE 1

- **Description of the example**

The e-beam deflector of the single deflection stage is to be used for pattern writing.

- **Description of the command**

The DEFMODE command specifies the e-beam deflection stage(s) (single or double) for pattern writing. The single stage e-beam deflection can perform more accurate pattern writing, while the double stage e-beam deflection can perform pattern writing in a shorter time. When this command is omitted, the double stage e-beam deflection will be applied.

① m

Specifies the e-beam deflection stage(s) using a number (1 for single-stage or 2 for double-stage) for pattern writing.

4.3.9 RESTOR

(Optional; applicable only in JBX-9300FS, JBX-6000FS, JBX-6000FS/E, JBX-6300 series and JBX-5500 series)

Format:

RESTOR [m] [sw,a]
① ② ③

Example:

RESTOR 1 (1)
RESTOR 1,CS,5.0 (2)

- **Description of the example**

- (1) This command line sets the electron optical system (EOS) parameters such as the distortion correction amount, the dynamic focus value, and the parameters for the lenses of the EOS to the values you previously saved, before writing starts. At this time, each lens of the EOS is demagnetized.
- (2) If the beam current, measured after the setting of the EOS parameters and the demagnetization of each lens, has changed by 5% or more compared to the beam current measured when the EOS parameter values were saved, the EOS column is automatically aligned. At this time, the current is adjusted so as to reduce the change in the current to 5% or less; then adjusting correction of the objective lens are automatically adjusted.

✎ Automatic EOS column alignment is a feature specific to the JBX-9300FS and JBX-6300 series. For details, see the manual, “Operations of Column Adjustment Program.”

● Description of the command

As a preliminary stage of pattern writing, the RESTOR command sets the parameters such as the distortion correction amount, the dynamic focus value, and the parameters for the EOS lenses to the values you previously saved. You can specify whether or not to demagnetize the EOS lenses at this time. Also, you specify how to align the EOS column automatically. Here you specify the maximum allowable change between the current measured after parameter setting and the current measured at the time of parameter saving. If you omit this command, the e-beam machine does not change these parameter settings but writes patterns using the parameter settings it already has at that time.

① m

Specifies whether or not to demagnetize all the lenses of the EOS will be.

m	Meaning
0	Do not demagnetize the EOS lenses of the e-beam machine.
1	Demagnetize the EOS lenses of the e-beam machine. (Default)

② sw

This specifies how to align the column automatically. If you omit this parameter, the e-beam machine does not align the EOS column automatically.

sw	Contents of automatic alignment
AL	The e-beam machine adjusts only the alignment (AL).
CF	After adjusting the alignment (AL) and the current, the e-beam machine carries out automatic adjustment (SFOCUS) of the objective lens focus (FOCUS).
CS	After adjusting the alignment (AL) and the current, the e-beam machine carries out automatic adjustment (SFOCUS) of the objective lens focus (FOCUS) and astigmatism (STIG) correction.

✎ The JBX-6000FS, JBX-6000FS/E and JBX-5500 series do not allow you to specify the parameter sw.

③ a

This specifies the maximum allowable change between the current measured at the time of EOS parameter saving and the current measured after EOS parameter setting, in percentage. If you specify CF or CS for the parameter sw, the e-beam machine adjusts the current during automatic column alignment to make the change of the current less than the value you specify here.

✎ If you specify the parameter sw, you cannot omit this parameter.

✎ The JBX-6000FS, JBX-6000FS/E and JBX-5500 series do not allow you to specify the parameter a.

4.3.10 OFFSET

(Optional)

Format:	
OFFSET	(dx,dy) ① ②
Example:	
OFFSET	(5000.0,-5000.0)

- **Description of the example**

The writing positions of patterns such as chips and character strings are uniformly shifted by (given a parallel displacement of) 5000.0 μm rightward and 5000.0 μm downward from the substrate center.

- **Description of the command**

The OFFSET command specifies the position displacement (offset) of the origin of the substrate center coordinate system defined in the jobdeck from the physical substrate center. The offset amount is applied to all of the parameters that are specified by the substrate center coordinate system in the jobdeck. To specify the offset amount, designate the distance from the origin of the substrate center coordinate system in the jobdeck, using the coordinate system where the origin is the physical substrate center. Positive increments indicate rightward displacement along the X-axis and upward displacement along the Y-axis.

- ① dx

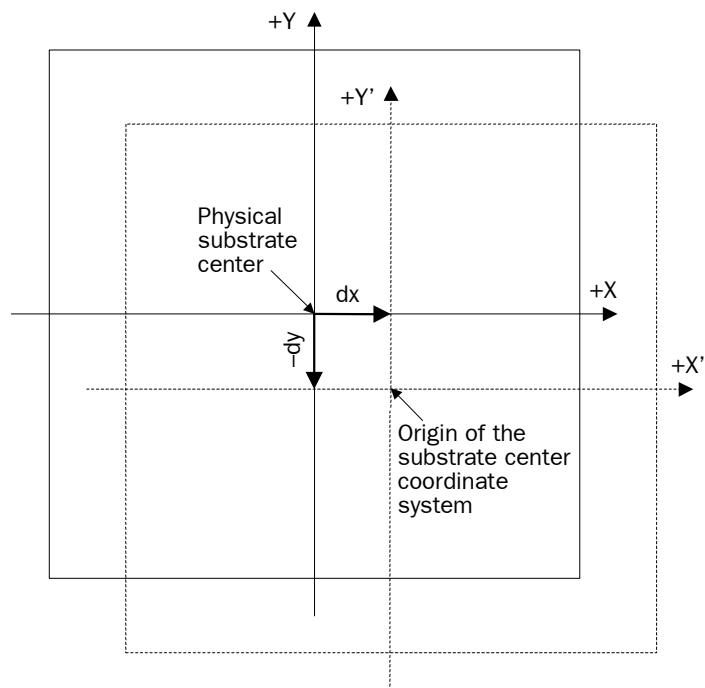
Specifies the X-coordinate to be given to the origin of the substrate center coordinate system.

- ② dy

Specifies the Y-coordinate to be given to the origin of the substrate center coordinate system.



The position relationship between the physical substrate center and the origin of the substrate center coordinate system is shown below.



4.3.11 MCOR

(Optional; inapplicable in the direct-writing mode)

Format:

MCOR gx,gy,ry
 ① ② ③

Example:

MCOR -0.5,0.5,1.2

● Description of the example

The writing positions of patterns such as chips and character string are uniformly expanded or contracted, and rotated, from the substrate center by reducing $0.5 \mu\text{m}$ per 100 mm along the X coordinate and by increasing $0.5 \mu\text{m}$ per 100 mm along the Y coordinate, and by displacing $1.2 \mu\text{m}$ rightward along the X coordinate per 100 mm increment along the Y coordinate.

● Description of the command

This command specifies the fine expansion or contraction correction amounts of the X coordinate axis and the Y coordinate axis, and the fine rotation correction amount of the Y coordinate axis, respectively, using the substrate center coordinate system specified in the jobdeck. These are generically called the orthogonal correction amounts of the substrate stage. All the parameters specified using the substrate center coordinate system in the jobdeck have expansion or contraction and rotation applied to them.

For the expansion or contraction amounts, designate offset values from the coordinate origin in μm per 100 mm for both X and Y axes using the substrate center coordinate system. Positive increments indicate rightward along the X-axis, and upward along the

Y-axis. For the rotation amount, designate an offset value along the X coordinate from the coordinate origin in μm per 100 mm along the Y coordinate.

① **gx**

Specifies the expansion or contraction amount along the X coordinate axis of the substrate center coordinate system.

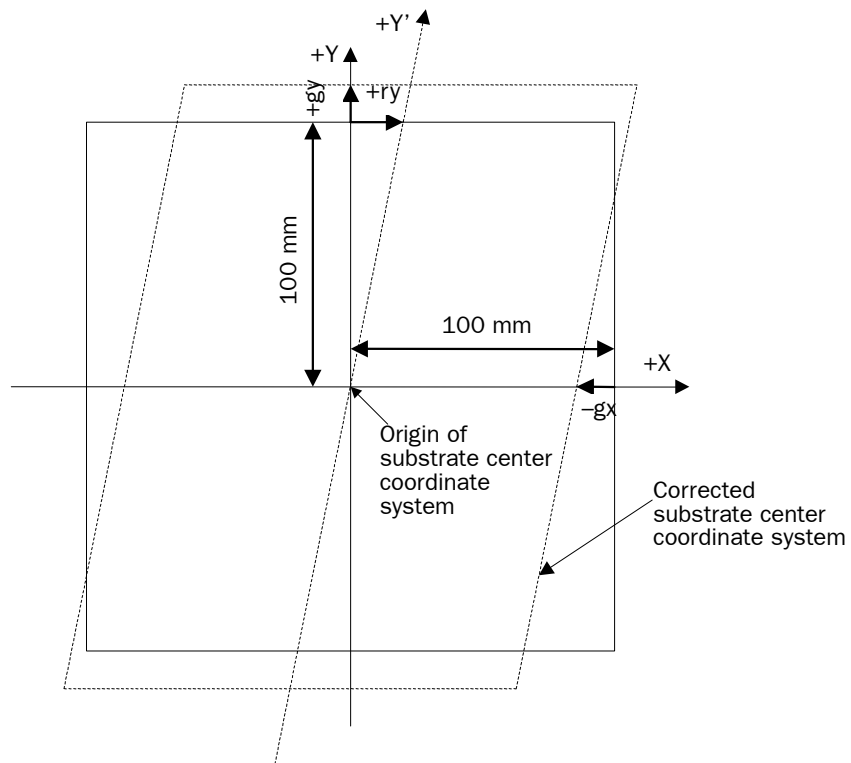
② **gy**

Specifies the expansion or contraction amount along the Y coordinate axis of the substrate center coordinate system.

③ **ry**

Specifies the rotation amount of the Y coordinate axis of the substrate center coordinate system.

 The definition of the orthogonal correction amounts of the substrate stage is shown below.



4.3.12 HEIGHT

(Optional)

Format:	
HEIGHT	f,gx,gy,rx,ry [,sh, sy] ①②③④⑤⑥⑦
Example:	
HEIGHT	19,64,-3,-1,0,1,-2

● Description of the example

In this example, the values below are assigned to the parameters of the HEIGHT command. These correct the amplitude and rotation of the writing field and the focus of the beam for the height difference between the substrate surface and the reference mark surface. Here, a value of 19 is given for the excitation current coefficient fine correction for the objective lens. For the deflection control voltage coefficient fine correction, a value of 64 is given for the X amplitude fine correction and -3 for the Y amplitude fine correction. Also, a value of -1 for the X rotation fine correction and 0 for the Y rotation fine correction are given. In order to compensate for beam shifts caused by the change of the excitation current, 1 for the X shift fine correction and -2 for the Y shift fine correction are added to each of the correction values.

● Description of the command

The HEIGHT command is used to correct the amplitude and rotation of the writing field and the defocusing of the beam caused by the height difference between the substrate surface and the reference mark surface. The fine correction values for the focusing lens excitation current coefficient and each deflector control voltage coefficient are specified. Also, the beam shift fine corrections due to the change of excitation current are specified.

① f

Specifies the fine correction value of the excitation current coefficient for the focus lens.

② gx

Specifies the X amplitude fine correction value of the control voltage coefficient for the e-beam deflection system.

③ gy

Specifies the Y amplitude fine correction value of the control voltage coefficient for the e-beam deflection system.

④ rx

Specifies the X rotation fine correction value of the control voltage coefficient for the e-beam deflection system.

⑤ ry

Specifies the Y rotation fine correction value of the control voltage coefficient for the e-beam deflection system.

⑥ sx

Specifies the X shift fine correction value for the beam shift caused by the change in the exciting current. Unless specified, the value is 0.

⑦ sy

Specifies the Y shift fine correction value for the beam shift caused by the change in the exciting current. Unless specified, the value is 0.

☞ For information about Each correction, refer to “Lithography Operations” (or “Operation”) for the lithography system.

4.3.13 SUBHEI

(Optional; applicable only in JBX-9300FS, JBX-6300 series and JBX-5500 series)

Format:

SUBHEI gx, gy, rx, ry, shx, shy
 ① ② ③ ④ ⑤ ⑥

Example:

SUBHEI 15, 18, -32, -35, 24, -18

● Description of the example

In this example, the values below are assigned to the parameters of the SUBHEI command. These correct the amplitude and rotation of the field of the deflector and the beam shift of the main deflector for the height difference between the substrate surface and the reference mark surface. Here, a value of 15 is given for the subsidiary deflector X amplitude control voltage coefficient fine correction and 18 for the Y amplitude fine correction. Also, a value of -32 for the X rotation fine correction and -35 for the Y rotation fine correction are given. In order to compensate for the beam shifts, 24 for the X shift fine correction and -18 for the Y shift fine correction are added to each of the correction values.

● Description of the command

The SUBHEI command is used to correct the amplitude and rotation of the field of the deflector and the beam shift of the main deflector for the height difference between the substrate surface and the reference mark surface. The fine correction values for each deflector control voltage coefficient are specified. Also, the beam shift fine corrections are specified.

① gx

Specifies the X amplitude fine correction value of the control voltage coefficient for the subsidiary deflection system.

② gy

Specifies the Y amplitude fine correction value of the control voltage coefficient for the subsidiary deflection system.

③ rx

Specifies the X rotation fine correction value of the control voltage coefficient for the subsidiary deflection system.

④ ry

Specifies the Y rotation fine correction value of the control voltage coefficient for the subsidiary deflection system.

- ⑤ shx
Specifies the X shift fine correction value for the beam shifting when the exciting current is changed.
- ⑥ shy
Specifies the Y shift fine correction value for the beam shifting when the exciting current is changed.
- ☞ For information about Each correction, refer to “Lithography Operations” (or “Operation”) for the lithography system.

4.3.14 SHTCOR

(Optional)

Format:	
SHTCOR	{ { [LINE=l] / [AREA=a] } { [RECT=r]
	① ② ③
–	/ [TRAPX=tx] / [TRAPY=ty] } }
	④ ⑤
Examples:	
SHTCOR	LINE=10.0, AREA=–10.0 (1)
SHTCOR	TRAPX=5.0, TRAPY=10.0 (2)

● Descriptions of the examples

- (1) Adds 10.0% of the shot time for single line writing, and –10.0% of that for area pattern writing, to each of the initial shot times when pattern writing is carried out.
- (2) Adds 5.0% of the shot time for X-trapezoid pattern writing, and 10.0% of that for Y-trapezoid pattern writing, to each of the initial shot times.

● Description of command

- ① l
Specifies the shot time correction amount for single line patterns in the spot e-beam machine.
- ② a
Specifies the shot time correction amount for area patterns in the spot e-beam machine.
- ③ r
Specifies the shot time correction amount for rectangular patterns in the variable shaped e-beam machine.
- ④ tx
Specifies the shot time correction amount for X-trapezoid patterns in the variable shaped e-beam machine.
- ⑤ ty
Specifies the shot time correction amount for Y-trapezoid patterns in the variable shaped e-beam machine.

The shot time for writing is obtained, as shown below, using the shot time (Stmod) and the shot time correction amount (Vc) that the command SHTCOR of this section adds; the shot time (Stmod) is obtained by adding the shot time modulation amount to the shot time that is specified in the jobdeck file or is calculated from items such as the beam current and the resist sensitivity or others.

$$(\text{Shot time for writing}) = \text{St mod} \times \frac{100 + \text{Vc}}{100}$$

 Stmod is the shot time to which the shot time modulation specified by the command MODULAT is added ( Sect. 4.3.47).

4.3.15 STEPPER

(Optional; applicable only in JBX-9000MV)

Format:	
STEPPER	's' ①
Example:	
STEPPER	'stepper_01'

● Description of the example

The stepper distortion correction function is to be used in the pattern writing routine according to the correction amounts stored in the stepper distortion correction file “stepper_01.stp.”

● Description of the command

The STEPPER command specifies the name of a file containing stepper distortion correction amounts (without the file type “.stp”). When this command is omitted, the stepper distortion correction function, which corrects the distortion specific to the stepper for making mask or reticle patterns, will be inactive during pattern writing.

① s

Specifies the name of a stepper distortion correction file.

4.3.16 MMDPRM

(Optional; applicable only in JBX-9300FS and JBX-6300 series)

Format:	
MMDPRM	'm' ①
Example:	
MMDPRM	'mmdcorr_01'

● Description of the example

The mix-and-match distortion correction function is to be used in the pattern writing routine according to the correction amounts stored in the mix-and-match distortion correction file “mmdcorr_01.mmd.”

- **Description of the command**

The MMDPRM command specifies the execution of pattern writing while adjusting the distortion specific to the stepper (mix-and-match distortion) when you implement mix-and-match pattern writing with a stepper for mask writing or for reticle writing. Specify, as the parameter, the name of the file containing various parameters for correcting the mix-and-match distortion (without the file type “.mmd”). When this command is omitted, the mix-and-match distortion correction function will be inactive during pattern writing.

① m

Specifies the name of the mix-and-match distortion correction file.

4.3.17 SHTOFS

(Optional; inapplicable in JBX-6000FS)

Format:

SHTOFS 'st'
 ①

Example:

SHTOFS 'shot_offset'

- **Description of the example**

The shot time offset function is to be used in the pattern writing routine according to the correction amounts stored in the shot time offset table file “shot_offset.sto.”

- **Description of the command**

The SHTOFS command specifies the name of a file containing shot time offset table (without the file type “.sto”). When this command is omitted, the shot time offset function that corrects for the dose amount change corresponding to beam sizes* will be inactive during pattern writing.

① st

Specifies the name of the shot time offset table file.

* This is for the JBX-9000MV variable shaped e-beam lithography system. For spot e-beam lithography systems such as the JBX-9300FS, this beam size is the finally partitioned pattern size for writing.

4.3.18 SPCPRM

(Optional: available only in JBX-3030MV and later models)

Format:	
SPCPRM	'spc' ①
Example:	
SPCPRM	'shot_position_corr'

- **Description of the example**

The system corrects the shot position during writing, according to the parameter settings contained in the parameter file “shot_position_corr” for shot-position correction for each shot size.

- **Description of the command**

The SPCPRM command specifies the shot-position correction writing for each shot size, and specifies the filename (without “.spc”) of the file that contains the parameter settings for shot-position correction for each shot size. The shot-position correction writing for each shot size is a feature that writes patterns while correcting the shot position according to the shot size. If this command is omitted, the default parameter settings specific to the lithography system take effect.

- ① **spc**

The parameter spc specifies the shot-size correction parameter filename.

-  The shot-position correction writing for each shot size is an optional feature for the JBX-3030MV and is a standard feature for the JBX-3040MV and later models. For details, see the manual, “General Statement” or the manual, “Operation” for your lithography system.

4.3.19 PECPRM

(Optional: available only in JBX-9000MV and JBX-3030MV and later models)

Format:	
PECPRM	'pec' ①
Example:	
PECPRM	'prox_effect_corr'

- **Description of the example**

The function for proximity effect correction writing is to be used in the pattern writing routine according to the parameters stored in the proximity effect correction parameter file prox_effect_corr.pec.

- **Description of the command**

This command specifies the name of the file containing proximity effect correction parameters (without the file type “.pec”). When this command is omitted, the default parameters specific to the e-beam lithography system will be used for pattern writing.

- ① **pec**

- Specifies the name of a proximity effect correction parameter file.

- **Description of the Supplement**

This command is to be individually given in each writing path definition, in which you wish to perform the proximity effect correction writing, in the jobdeck file.

☞ The proximity-effect correction writing is an optional feature for the JBX-9000MV and the JBX-3030MV. For details, see the manual, “General Statement,” or the manual, “Lithography Operations” (for the JBX-9000MV) or “Operation” (for the JBX-3030MV) for your lithography system.

4.3.20 FOGPRM

(Optional: available only in JBX-9000MV and JBX-3030MV and later models)

Format:

FOGPRM 'fog'
 ①

Example:

FOGPRM 'fogging'

- **Description of the example**

The function for fogging error correction writing is to be used in the pattern writing routine according to the parameters stored in the fogging error correction parameter file fogging.fed.

- **Description of the command**

This command specifies execution of fogging error correction writing and also the name of the file containing fogging error correction parameters (without the file type “.fed”). When this command is omitted, the fogging error correction will not be carried out during pattern writing.

- ① **fog**

- Specifies the name of a fogging error correction parameter file.

4.3.21 PRCPRM

(Optional: available only in JBX-9000MV and JBX-3030MV and later models)

Format:	
PRCPRM	'prc' ①
Example:	
PRCPRM	'process'

- **Description of the example**

The function for process error correction writing is to be used in the pattern writing routine according to the parameters stored in the process error correction parameter file process.prc.

- **Description of the command**

This command specifies execution of process error correction writing and also the name of the file containing process error correction parameters (without the file type “.prc”). When this command is omitted, the process error correction will not be carried out during pattern writing.

① prc

Specifies the name of a process error correction parameter file.

4.3.22 PRJPRM

(Optional: available only in JBX-9000MV and JBX-3030MV and later models)

Format:	
PRJPRM	'prj' ①
Example:	
PRJPRM	'projection'

- **Description of the example**

The function for projection error correction writing is to be used in the pattern writing routine according to the parameters stored in the projection error correction parameter file projection.prc.

- **Description of the command**

This command specifies execution of projection error correction writing and also the name of the file containing projection error correction parameters (without the file type “.prj”). When this command is omitted, the projection error correction will not be carried out during pattern writing.

① prj

Specifies the name of a projection error correction parameter file

4.3.23 LECPRM

(Optional: available only in JBX-3030MV and later models)

Format:	
LECPRM	'lec' ①
Example:	
LECPRM	'micro_loading'

- **Description of the example**

The system corrects for the middle range effect during writing, according to the parameter settings contained in the middle range-effect correction parameter file “micro_loading.lec.”

- **Description of the command**

The LECPRM command specifies the middle range-effect correction writing, and specifies the filename (without “.lec”) of the file that contains the middle range-effect correction parameter settings. If this command is omitted, the system does not correct for the middle range effect during writing.

① lec

The parameter lec specifies the middle range-effect correction parameter filename.

☞ The middle range-effect correction writing is an optional feature for the JBX-3030MV. For details, see the manual, “General Statement” or the manual, “Operation” for your lithography system.

4.3.24 GLECPRM

(Optional: available only in JBX-3040MV and later models)

Format:	
GLECPRM	'gl' ①
Example:	
GLECPRM	'global_loading'

- **Description of the example**

The system corrects for the global-loading effect during writing, according to the parameter settings contained in the global-loading effect correction parameter file “global_loading.glec.”

- **Description of the command**

The GLECPRM command specifies the global-loading effect correction writing, and specifies the filename (without “.glec”) of the file that contains the global-loading effect correction parameter settings. If this command is omitted, the system does not correct for the global-loading effect during writing.

① gl

The parameter gl specifies the global-loading effect correction parameter filename.

4.3.25 ANG2SIZ

(Optional: available only in JBX-3030MV and later models)

Format:	
ANG2SIZ	'a2s'
	①
Example:	
ANG2SIZ	'angle'

● Description of the example

The function for trapezoid dimensional correction writing at any of the angles is to be used in the pattern writing routine according to the parameters stored in the trapezoid dimensional correction parameter file “angle.a2s”.

● Description of the command

This command specifies execution of trapezoid dimensional correction writing in accordance with angles of the oblique sides of the trapezoid and also the name of the file containing trapezoid dimensional correction parameters (without the file type “.a2s”). When this command is omitted, the correction will not be carried out during pattern writing.

① a2s

Specifies the name of a trapezoid dimensional correction parameter file

4.3.26 SSCPRM

(Optional: available only in JBX-3030MV and later models)

Format:	
SSCPRM	'ssc'
	①
Example:	
SSCPRM	'shot_size_corr'

● Description of the example

The system corrects the shot size during writing, according to the parameter settings contained in the shot-size correction parameter file “shot_size_corr.”

- **Description of the command**

The SSCPRM command specifies to carry out the shot-size correction writing for each position in the field plane, and specifies the filename (without “.ssc”) of the file that contains the shot-size correction parameter settings. The shot-size correction writing for each position in the field plane is a feature that writes patterns according to the position in the field while correcting the size and position of the electron beam’s shot shapes that make up the pattern data. If this command is omitted, the system does not correct the shot size during writing.

- ① **ssc**

The parameter ssc specifies the shot-size correction parameter filename.

☞ The shot-size correction writing for each position in the field plane is an optional feature for the JBX-3030MV and a standard feature for the JBX-3040MV and later models. For details, see the manual, “General Statement” or the manual, “Operation” for your lithography system.

4.3.27 PEDPRM

(Optional: available only in JBX-3050MV and later models)

Format:	
PEDPRM	'ped' ①
Example:	
PEDPRM	'postexpdelay'

- **Description of the example**

According to the parameters contained in the time-dependent dose-correction parameter file “postexpdelay.ped”, the system performs time-dependent dose correction during writing.

- **Description of the command**

The PEDPRM command specifies the execution of the time-dependent dose-correction writing, and also specifies the file name (without “.ped”) of the file containing the time-dependent dose-correction parameters. With this command omitted, the system does not perform time-dependent dose correction during writing.

- ① **ped**

Here you specify the time-dependent dose-correction parameter file name.

4.3.28 WARMUP

(Optional)

Format:
WARMUP t[,m]
① ②

Example:
WARMUP 5.0

● Description of the example

This example designates a five minute delay for starting e-beam writing after loading a substrate onto the stage.

● Description of the command

This command specifies a delay (waiting) time after loading a substrate onto the stage. It also specifies an operation mode, called a conditioning mode, of the e-beam machine during the delay time so as to stabilize the temperature of the substrate and the electron optics system of the e-beam machine.

① t

Specifies the delay time for starting pattern writing after loading a substrate onto the stage. However, when you specify dummy pattern writing (m = 2) as the operation mode ②, the delay time will terminate when the temperature fluctuation amount and the beam position fluctuation amount that are to be periodically measured have converged within the tolerances specified in various system correction parameters*.

② m**

Specifies the operation mode (conditioning mode) of the e-beam machine during the delay time.

m	Conditioning mode
0	Continues switching the e-beam on and off with the interval specified by various system correction parameters. This is the default value for the JEOL variable shaped e-beam machine.
1	Keeps the e-beam on. This is the default value for the JEOL spot e-beam machine. For the variable shaped e-beam machine, the beam size will also be varied according to various system correction parameters.
2	Continuously executes the dummy pattern writing using predetermined pattern data at the position specific to the e-beam machine. During that time, the temperature fluctuation of the e-beam machine and the position fluctuation of the e-beam are periodically measured according to the various system correction parameters. When these fluctuations converge within the tolerances before the end of the specified delay time t the system will terminate this WARMUP routine and start the actual pattern writing.

* These parameters are set in the calibration program and are specified using the EOS command and/or CALPRM command.

** In the case of JBX-6000FS, this parameter is inapplicable. Therefore, even if you specify m = 0 or m = 2, m = 1 will be set.

4.3.29 LBC

(Optional; inapplicable in JBX-9000MV and JBX-3030MV)

Format:	
LBC	sw ①
Example:	
LBC	HOLD

- **Description of the example**

The operation mode of the LBC is set to HOLD. That is, e-beam writing is executed without using the LBC (laser beam control), which corrects the e-beam position.

- **Description of the command**

This command specifies whether or not to activate the LBC. The LBC is used to correct the position error of the stage by deflecting the e-beam. When the LBC is activated, the position of the stage is monitored at all times during e-beam writing, and the position error of the stage is automatically corrected for. When this command is omitted, the LBC is activated during e-beam writing.

- ① sw

Specifies the operation mode of the LBC.

sw	Operation mode
ON	LBC is activated during e-beam writing. (Default)
HOLD	LBC is disabled during e-beam writing.

4.3.30 HSWITCH

(Optional for direct writing; applies only to JBX-9300FS and JBX-6300 series)

Format:

HSWITCH swg swc
 ① ②

Example:

HSWITCH OFF, OFF

- **Description of the example**

The substrate height measurement function at the global mark positions and the chip mark positions is to be off during direct writing.

- **Description of the command**

The HSWITCH command specifies the substrate height measurement function mode for direct writing. The substrate height measurement function is used to measure optically the height of the substrate surface at the positions of global marks and chip marks on the substrate. When the substrate height measurement function is activated in direct writing, the substrate height measurement is executed during global mark and chip mark detection at their positions, and the pattern writing correction is made according to that result. When this command is omitted, the substrate measurement function is activated.

① swg

Specifies whether or not to activate the substrate height measurement function at the global mark detection position.

swg	Operation mode
ON	The substrate height measurement function is activated. (Default)
OFF	The substrate height measurement function is disabled.

② swc

Specifies whether or not to activate the substrate height measurement function at the chip mark detection position.

swc	Operation mode
ON	The substrate height measurement function is activated. (Default)
OFF	The substrate height measurement function is disabled.

4.3.31 GLMPRT

(Optional for both direct writing and phase shift mask writing)

Format:	
GLMPRT	ps[,fw[,m]]
	① ② ③
Example:	
GLMPRT	230.0,100.0

● **Description of the example**

The e-beam irradiation is to be made on the mark protection area that consists of the mark detection area and a frame around it having a width of 100.0 μm in all four directions (up, down, left, and right). The irradiation begins after global mark detection by default, and continues until the e-beam dose reaches 230.0 $\mu\text{C}/\text{cm}^2$.

● **Description of the command**

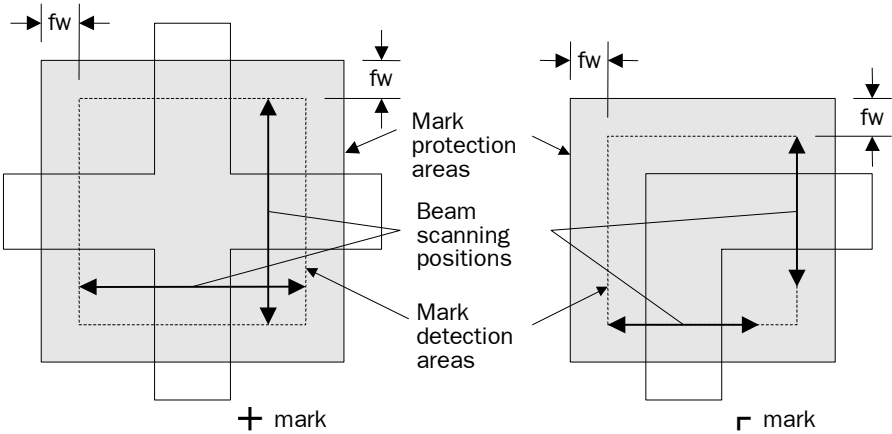
The GLMPRT command specifies the e-beam irradiation parameters for global-mark protection. For mark protection, the resist must be left on the mark so that the mark is not dissolved by subsequent processes. E-beam irradiation on the resist is to be carried out to keep the resist on the mark.

① ps

Specifies the total e-beam irradiation.

② fw

Specifies, as the extended mark protection area, the width of the frame area surrounding the mark detection area on all four directions (up, down, left, and right).



③ m

Specifies an operation mode for mark protection.

m	Execution mode for mark protection
0	Mark protection is performed after detecting the mark. (Default)

m	Execution mode for mark protection
1	Mark protection is performed before detecting the mark.

— CAUTION —

- For mark protection, the e-beam current or the e-beam current density must be measured during e-beam writing.
- To measure the e-beam current or the e-beam current density, you must either specify measurement of beam current using the CURRNT command in the AUTO CALIB menu that is performed during e-beam writing, or specify the SHOT command so that the writing time for each e-beam shot is to be determined from the e-beam current or the e-beam current density.

4.3.32 PLMPRT

(Optional for phase shift mask writing)

Format:	
PLMPRT	ps [,fw [,m]]
	① ② ③
Example:	
PLMPRT	230.0,100.0

● Description of the example

The e-beam irradiation is to be made on the mark protection area that consists of the mark detection area and a frame around it having a width of 100.0 μm in all four directions (up, down, left, and right). This irradiation begins after plate mark detection by default, and continues until the e-beam dose on that area reaches 230.0 $\mu\text{C}/\text{cm}^2$.

● Description of the command

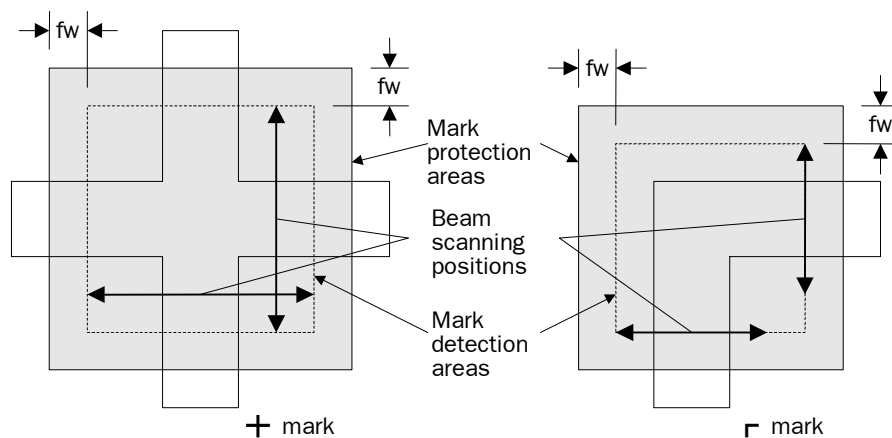
The PLMPRT command specifies the e-beam irradiation parameters for plate-mark protection. For mark protection, the resist must be left on the mark so that the mark is not dissolved by subsequent processes. E-beam irradiation on the resist is to be carried out to keep the resist on the mark.

① ps

Specifies the total e-beam irradiation.

② fw

Specifies, as the extended mark protection area, the width of the frame area surrounding the mark detection area in all four directions (up, down, left, and right).



③ m

Specifies an operation mode for mark protection.

m	Execution mode for mark protection
0	Mark protection is performed after detecting the mark. (Default)
1	Mark protection is performed before detecting the mark.

— CAUTION —

- For mark protection, the e-beam current or the e-beam current density must be measured during e-beam writing.
- To measure the e-beam current or the e-beam current density, you must either specify measurement of beam current using the CURRNT command in the AUTO CALIB menu that is performed during e-beam writing, or specify the SHOT command so that the writing time for each e-beam shot is to be determined from the e-beam current or the e-beam current density.

4.3.33 CHMPRT

(Optional for direct writing)

Format:
 CHMPRT ps[,fw[,m]]
 ① ② ③

Example:
 CHMPRT 230.0,100.0

● Description of the example

E-beam irradiation is to be made on the mark protection area that consists of the mark detection area and frame around it having a width of 100.0 μm in all four directions (up, down, left, and right). This irradiation begins after chip mark detection by default, and continues until the e-beam dose reaches 230.0 $\mu\text{C}/\text{cm}^2$.

● Description of the command

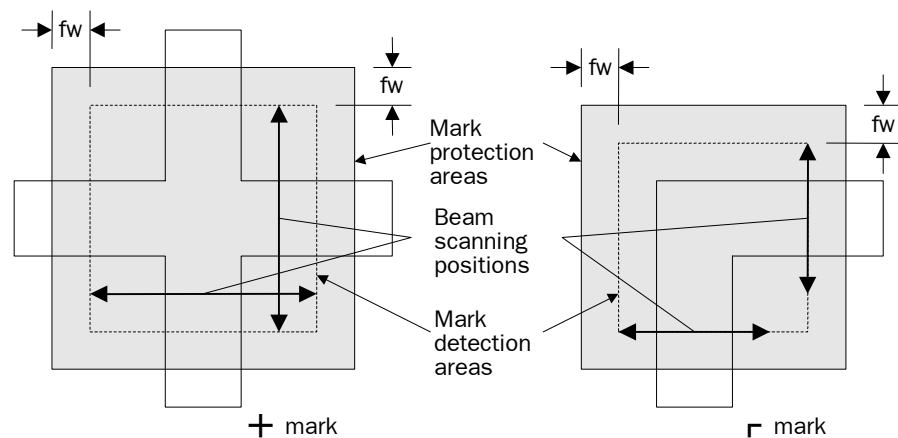
The CHMPRT command specifies the e-beam irradiation parameters for chip mark protection. For mark protection, the resist must be left on the mark so that the mark is not dissolved by subsequent processes. E-beam irradiation on the resist is to be carried out to keep the resist on the mark.

① ps

Specifies the total e-beam irradiation.

② fw

Specifies, as the extended mark protection area, the width of the frame area surrounding the mark detection area in all four directions (up, down, left, and right).



③ m

Specifies an operation mode for mark protection.

m	Execution mode for mark protection
0	Mark protection is performed after detecting the mark. (Default)
1	Mark protection is performed before detecting the mark.

— CAUTION —

- For mark protection, the e-beam current or the e-beam current density must be measured in the pattern writing routine.
- To measure the e-beam current or the e-beam current density, you must either specify measurement of beam current using the CURRNT command in the AUTO CALIB menu that is performed during e-beam writing, or specify the SHOT command so that writing time for each e-beam shot is to be determined from the e-beam current or the e-beam current density.

4.3.34 GLMDET

(Optional for direct writing and for phase shift mask writing)

Format:

GLMDET sw
 ①

Example:

GLMDET A

● Description of the example

The writing control program (EXPOSE) automatically carries out global mark detection. When mark detection fails, the system will skip pattern writing on that substrate and proceed to the pattern writing that follows in the schedule.

● Description of the command

The GLMDET command specifies the operation mode of the writing control program (EXPOSE) for mark detection of the global mark. When this command is omitted, the program operates in the AUTO mode.

① sw

Specifies the operation mode of the writing control program (EXPOSE) for mark detection of the global mark.

sw	Operation mode
A (AUTO)	The writing control program (EXPOSE) automatically implements mark detection, according to the contents of the jobdeck file, the schedule file, and the CALIB parameter file, which contains various correction parameters*. When mark detection of a substrate fails, the system skips pattern writing on that substrate and proceeds to the pattern writing that follows in the schedule. (Default)
S (SEMI)	The writing control program (EXPOSE) automatically implements mark detection, according to the contents of the jobdeck file, the schedule file, and the CALIB parameter file, which contains various correction parameters. When mark detection fails, the system asks the operator about the operations that follow.  This function S (SEMI) is inapplicable at present.
M (MANUAL)	The operator manually carries out mark detection using the procedure that is equivalent to that of the AUTO mode.
C (CANCEL)	No global mark detection is carried out.

* These parameters are set in the calibration program and specified using the EOS command and/or CALPRM command.

4.3.35 PLMDET

(Optional for phase shift mask writing)

Format:	
PLMDET	sw ①
Example:	
PLMDET	A

- **Description of the example**

The writing control program (EXPOSE) automatically carries out the detection of the plate mark. When mark detection fails, the system skips pattern writing on that substrate and proceeds to the pattern writing that follows in the schedule.

- **Description of the command**

The PLMDET command specifies the operation mode of the writing control program (EXPOSE) for mark detection of the plate mark. When this command is omitted, the program operates in the AUTO mode.

① **sw**

Specifies the operation mode of the writing control program (EXPOSE) for mark detection of the plate mark.

sw	Operation mode
A (AUTO)	The writing control program (EXPOSE) automatically implements mark detection, according to the contents of the jobdeck file, the schedule file, and the CALIB parameter file, which contains various correction parameters*. When mark detection of a substrate fails, the system skips pattern writing on that substrate and proceeds to the pattern writing that follows in the schedule. (Default)
S (SEMI)**	The writing control program (EXPOSE) automatically implements mark detection, according to the contents of the jobdeck file, the schedule file, and the CALIB parameter file, which contains various correction parameters. When mark detection fails, the system asks the operator about the operations that follow.

* These parameters are set in the calibration program and specified using the EOS command and/or CALRM command.

** The S (SEMI) function is inapplicable at present.

4.3.36 CHMDET

(Optional for direct writing)

Format:

CHMDET sw [①, [e] ②, [n] ③]

Example:

CHMDET A

● Description of the example

The writing control program (EXPOSE) automatically carries out mark detection of the chip mark and height measurement at the chip mark position. When mark detection height measurement fails, the program skips pattern writing of the block of chips whose writing position depends on that detection by default. Also, when the chip mark detections and height measurements for the first and the second blocks of chips fail successively on the substrate, the program skips the pattern writing on that substrate and proceeds to the pattern writing that follows in the schedule. (The program operation for two successive failures from the beginning, which is described above, is the default of this command.)

● Description of the command

The CHMDET command specifies the operation mode of the writing control program for mark detection height measurement of the chip marks. When this command is omitted, the program operates in the AUTO mode.

① sw

Specifies the operation mode of the writing control program for chip mark detection and height measurement.

sw	Operation mode
A (AUTO)	The writing control program (EXPOSE) automatically implements mark detection height measurement, according to the contents of the jobdeck file, the schedule file, and the CALIB parameter file, which contains various correction parameters*. When mark detection or height measurement of a substrate fails, program operation is carried out in accordance with the values given in the following ② and ③. (Default)
S (SEMI)**	The writing control program (EXPOSE) automatically implements mark detection height measurement, according to the contents of the jobdeck file, the schedule file, and the CALIB parameter file, which contains various correction parameters. When mark detection or height measurement fails, the program asks the operator about the operations that follow.

② e

This parameter specifies whether or not to execute pattern writing of the block of chips that depends on the chip mark detection or height measurement when that

* These parameters are set in the calibration program and specified using the EOS command and/or CALRM command.

** The function S (SEMI) is inapplicable at present.

chip mark detection or height measurement fails. The parameter e is in effect only when A (AUTO) is specified in ①. When this parameter is not specified, it is set to 0 (default).

e	Whether or not to execute chip writing
0	Does not execute chip writing. (Default)
1	Executes chip writing.

③ n

This parameter specifies the upper limit of the number of consecutive failures of the chip mark detection or height measurement, counted from the first chip mark detection or height measurement. If the number of chip mark detection or height measurement failures reaches the upper limit value specified by this parameter, the program skips pattern writing for the corresponding substrate and carries out the pattern writing that follows in the schedule. This parameter is in effect only when A (AUTO) is specified in ①. When the parameter n is not specified, it is set to 2 (default).

4.3.37 WFR3M

(Optional for direct writing; applicable only in JBX-6000FS, JBX-6300 series and JBX-5500 series)

Format:	
WFR3M	m
	①
Example:	
WFR3M	1

● Description of the example

Chip mark detection is carried out only for the M1 mark, and only the position displacement (the amount of shift) of the chip mark center is corrected for in pattern writing.

● Description of the command

The WFR3M command specifies the mark detection mode for the chip mark and the writing correction mode. When this command is omitted, the mode is set to 3 (default).

① m

Specifies the mark detection mode of the chip mark and the writing correction mode. Designate a number (1 through 9) to specify the mode.

m	Mark detection mode and writing correction mode
1	Detects the M1 mark and corrects for the position displacement of the chip mark center.
2	Detects the M1, M2, and M3 marks; then corrects for the position displacement of the chip mark center, and the size (amplitude) and the

m	Mark detection mode and writing correction mode
	rotation of the beam deflection area.
3	Detects the M1, M2, and M3 marks; then corrects for the position displacement of the chip mark center, and the size (amplitude) of the beam deflection area. (Default)
4	Optional.*
5	Optional.
6	Detects the M1 mark; then corrects for the position displacement (shift amount) of the chip-mark center and corrects the size (amplitude) and rotation of the beam-deflection area. (Optional for JBX-5000LS, JBX-5000LS/E, JBX-6000FS, JBX-6000FS/E and JBX-5500 series)
7	Detects the M1 mark; then corrects for the position displacement (shift amount) of the chip-mark center and corrects the size (amplitude) of the beam-deflection area. (Optional for JBX-5000LS, JBX-5000LS/E, JBX-6000FS, JBX-6000FS/E and JBX-5500 series)
8	Switches the instrument to the SEM image observation mode. The operator can move the workstage and adjust the mark center of the M1 chip mark to the center of the SEM image. The stage position is to be set to the center of the M1 mark.
9	Detection of the M1, M2, and M3 marks is not carried out in this mode. Corrections are made to the position displacement of the chip mark center, and the size (amplitude) and the rotation of the beam deflection area. These are carried out by using data based on the position displacement of the global mark, and expansion (or contraction) and rotation correction amount of the substrate which are obtained by global mark detection.

4.3.38 CHIPAL

(Optional for direct writing; inapplicable in JBX-6000FS)

Format:	
CHIPAL	m ①
Example:	
CHIPAL	1

● Description of the example

Chip mark detection is carried out only for the M1 mark, and only the position displacement (the amount of shift) of the chip mark center is corrected for in pattern writing.

● Description of the command

The CHIPAL command specifies the mark detection mode for the chip mark and the writing correction mode. When this command is omitted, mode 4 is used by default.

① m:

* Modes 4 to 7 are options for specific users.

Specifies the mode for chip mark detection mode and for writing correction. Designate a number or a character in the following table to specify the mode.

m	Mode for mark detection and writing correction
0	No chip mark detection is carried out in this mode. The position displacement of the chip mark center, and the size (amplitude) and the rotation of the beam deflection area will be corrected for using data based on the position displacement of the global mark, and expansion (or contraction) and rotation correction amount of the substrate obtained by global mark detection.
1	Detects the M1 chip mark and corrects for the position displacement of the chip mark center.
4	Detects the M1, M2, M3, and M4 chip marks; then corrects for the position displacement of the chip mark center, and the size (amplitude) and the rotation of the beam deflection area. (Default)
S	The operator finds the position of the M1 chip mark by observing the SEM image and corrects for the position displacement from the chip mark center.*
V1	Carries out height measurement at the M1 chip mark position without mark detection.
V4	Carries out height measurement at the M1, M2, M3, and M4 chip mark positions without mark detections.



Detection of the position of chip mark M1 by observation of the SEM image
Here is the procedure: switch the lithography system to the SEM-image observation mode; move the substrate stage to align the center of chip mark M1 with the center of the SEM image; then, the resulting stage position is regarded as the mark center of M1. For the JBX-6300 series, this feature requires the purchase of the “4-Point Mark-Detection System.” This is currently not available in the JBX-5500 series.

4.3.39 CDRIFT

(Optional)

Format:		
CDRIFT	a1	[, [a2] [,m]]
	①	② ③
Example:		
CDRIFT	10.0	

● Description of the example

If the program detects a beam current change of $\pm 10\%$ or more from the initial measured beam current, or of $\pm 10\%$ (default) or more from the last measured beam current during e-beam writing, the further execution of the writing control program (EXPOSE) will be stopped because the parameter m has the default value (A).

* The machine enters the SEM image observation mode. Set the center of the SEM image to the center of the M1 mark by moving the workstage. This set the stage position is set to the center of the M1 mark.

- **Description of the command**

The CDRIFT command specifies the tolerance for beam current change and the operation mode of the writing control program (EXPOSE) after beam current detection during e-beam writing. When no beam current measurement is carried out during e-beam writing, the specification by this command has no effect.

① a1

Specifies the tolerance of beam current change from the initial measured beam current during e-beam writing.

② a2

Specifies the tolerance of the beam current change from the last measured beam current carried during e-beam writing. When this parameter is omitted, the same value specified in ① is applied.

③ m

Specifies the operation mode of the writing control program (EXPOSE) when the detected beam current change exceeds the tolerance specified in either ① or ② above.

m	Operation of writing control program (EXPOSE)
A (ABORT)	Stops the pattern writing of the cassette block being written and of all the schedule that follows. The writing control program (EXPOSE) terminates. (Default)
S (SKIP)	Stops the pattern writing of the cassette block being written and performs the pattern writing in the schedule that follows.

4.3.40 PDRIFT

(Optional)

Format:	
PDRIFT	a1 [, [a2] [,m]]
	① ② ③
Example:	
PDRIFT	0.2

- **Description of the example**

If the system detects a beam position displacement of $\pm 0.2 \mu\text{m}$ or more from the initial measured beam position, or of $\pm 0.2 \mu\text{m}$ (default) or more from the last measured beam position during e-beam writing, the further execution of the writing control program (EXPOSE) will be stopped because the parameter m has the default value (A).

- **Description of the command**

The PDRIFT command specifies the tolerance for beam position displacement and the operation mode of the writing control program (EXPOSE) after beam position detection during e-beam writing. When no beam position measurement is carried out during e-beam writing, the specification by this command has no effect.

- ① a1
Specifies the tolerance of the beam position displacement from the initial measured beam position during e-beam writing.
- ② a2
Specifies the tolerance of the beam position displacement from the last measured beam position during e-beam writing. When this parameter is omitted, the same value specified in ① is applied.
- ③ m
Specifies the operation mode of the writing control program (EXPOSE) when the detected beam position displacement exceeds the tolerance specified in either ① or ② above.

m	Operation of writing control program (EXPOSE)
A (ABORT)	Stops the pattern writing of the cassette block being written and of all the schedule that follows. The writing control program (EXPOSE) terminates. (Default)
S (SKIP)	Stops the pattern writing of the cassette block being written; then performs the pattern writing in the schedule that follows.

4.3.41 TDRIFT

(Optional)

Format:	
TDRIFT	a1 [, [a2] [,m]] ① ② ③
Example:	
TDRIFT	0.02

● Description of the example

If the system detects a temperature change of $\pm 0.02^{\circ}\text{C}$ or more from the initial measured substrate temperature, or of $\pm 0.02^{\circ}\text{C}$ (default) or more from the last measured substrate temperature on the workstage during e-beam writing, the further execution of the writing control program (EXPOSE) will be stopped because the parameter m is set to the default value (A).

● Description of the command

The TDRIFT command specifies the tolerance for substrate temperature change and the operation mode of the writing control program (EXPOSE) after substrate temperature detection on the workstage during e-beam writing. When no substrate temperature measurement is carried out during pattern writing, the specification by this command has no effect.

- ① a1
Specifies the tolerance of substrate temperature change from the initial measured substrate temperature on the workstage during e-beam writing.

② a2

Specifies the tolerance of the substrate temperature change from the last measured substrate temperature on the workstage during e-beam writing. When this parameter is omitted, the value specified in ① is applied.

③ m

Specifies the operation mode of the writing control program (EXPOSE) when the detected substrate temperature change exceeds the tolerance specified in either ① or ② above.

m	Operation of writing control program (EXPOSE)
A (ABORT)	Stops the pattern writing of the cassette block being written and of all the schedule that follows. The writing control program (EXPOSE) terminates. (Default)
S (SKIP)	Stops the pattern writing of the cassette block being written; then performs the pattern writing in the schedule that follows.

4.3.42 PATH

(Optional)

Format:		
PATH (n)	[m]	
①	②	
or		
PATH (n)	['m']	(1)
Examples:		(2)
PATH (1)	MASK50	
PATH (2)		

● Descriptions of the examples

- (1) Various corrective measurements for pattern writing (CALIB) that are defined by the menu named "MASK50" are to be carried out in the e-beam writing path PATH (1).
- (2) No writing corrective measurements for pattern writing (CALIB) are to be executed in the e-beam writing path PATH (2). The correction values in the preceding e-beam writing path will continue to be in effect.

 The PATH number given to each e-beam writing path specifies the order of writing.

● Description of the command

The PATH command specifies whether or not to carry out various corrective measurements for pattern writing (CALIB) in each e-beam writing path. It also specifies the menu name of the CALIB to be carried out. When no menu name is specified in a path, no corrective measurements for pattern writing are carried out in that path, and the correction values for the pattern writing in the preceding path will continue to be in effect. When this command is omitted, the PATH name given in the jobdeck becomes the menu name of the CALIB. In this case, if no PATH name is written in the jobdeck, no

corrective measurements for pattern writing are carried out in that path, and the correction values in the preceding path will continue to be in effect.

① n

Specifies the PATH number (which specifies the order of writing in each e-beam writing path) in the jobdeck for pattern writing.

② m

Specifies the menu name of the various corrective measurements for pattern writing (CALIB), which are carried out in the e-beam writing path specified by ①. The menu name must be issued in the AUTO CALIB menu file. If you omit this parameter, no corrective measurements for pattern writing are carried out in that path. If the character string that indicates the menu name begins with a character other than a letter, enclose the entire character string in single quotes ('').*

4.3.43 P

(Optional)

Format:

P(i) 'p'[,m,n[,m',n']]

① ② ③ ④

Examples:

P (1) 'jeol.j5x' (1)

P (2) 'JEOL' (2)

P (3) 'jeol.j5x',MULTI,3 (3)

P (4) 'jeol.j5x',GATHER,3,OVERLAP,2 (4)

● Descriptions of the examples

- (1) The chip pattern stored in the file jeol.j5x will be written on the chips to which the chip pattern assigned by the pseudo pattern number 1 in the jobdeck belongs.
- (2) The chip pattern having the ID name JEOL will be written on the chips to which the chip pattern assigned by the pseudo pattern number 2 in the jobdeck belongs.
- (3) The chip pattern stored in the file jeol.j5x will be written using the triple field shifting overlap writing mode on the chips to which the chip pattern assigned by the pseudo pattern number 3 in the jobdeck has belonged.
- (4) The chip pattern stored in the file jeol.j5x will be not only written using the field combine mode that writes 3×3 fields as a block but also written using the double overlap writing mode at each field at which the chip pattern assigned by the pseudo pattern number 4 in the jobdeck belongs.

The writing position at each chip center is specified by the parameters (δx , δy) of the P command in the jobdeck file.

● Description of the command

The P command specifies both a chip pattern to be written and the special writing mode that corresponds to the pseudo pattern number assigned in the jobdeck. The chip pattern

* This description method is available only in the JBX-3030MV.

and special writing mode specified here to a pseudo pattern number override those specified in the jobdeck if they concern the same pseudo pattern number. However, the writing position at each chip center will be that specified by the parameters (δx , δy) of the P command in the jobdeck file.

① i

Specifies the pseudo pattern number for chip writing.

② p

Specifies the ID name of the chip pattern data for chip writing or the name of a file containing the chip pattern data for chip writing.

 For lithography systems controlled by a DEC (now an HP) VAX series computer system, such as the JBX-6000FS, this parameter specifies the pattern data ID name of the chip. For lithography systems controlled by a Compaq (now an HP) AlphaStation computer system, such as the JBX-9000MV, JBX-9300FS, JBX-3030MV, and JBX-6000FS/E, or by a Sun Solaris series computer system, such as the JBX-6300 series and the JBX-5500 series, this parameter specifies the name of the file that contains the chip pattern data.

③ m

Specifies the special writing mode.

m	Special writing method
MULTI	Specifies execution of the field shifting overlap writing mode*. In this mode, overlap pattern writing is performed n times, applying one nth of the normal e-beam dose, by changing the writing position on the writing field of pattern data .
MULTI/FP	Carries out the field shift overlap writing for the entire chip, in order from the first overlap, without mixing the writing fields of different overlaps.
GATHER	Specifies execution of the field combine writing mode**. In this mode, the patterns of up to $n \times n$ fields are handled as a single field block and written for pattern data of which the field size is one nth of the maximum field size.
OVERLAP	Specifies execution of overlap writing mode***. In this mode, overlap pattern writing is performed n times, applying one nth of the normal e-beam dose on the writing fields of pattern data.

 For information about field shift overlap writing, refer to “General Statement” or “Lithography Operations” (or “Operation”) for each lithography system other than JBX-6000FS.

 For information about field combine writing, refer to “General Statement” or “Lithography Operations” (or “Operation”) for each lithography system other than JBX-6000FS.

 For information about overlap writing, refer to “General Statement” or “Lithography Operations” (or “Operation”) for each lithography system other than JBX-6000FS.

 The MULTI/FP writing is called the full pass writing. You can specify the full pass writing only in the JBX-3050MV and later models.

You can specify GATHER and OVERLAP at the same time. Also, you can specify

* For details about field shift overlap writing, see the separate instruction manuals.

** For details about field combine writing, see the separate instruction manuals.

*** For details about overlap writing, see the separate instruction manuals.

MULTI and GATHER simultaneously. In these cases, specify one as m and the other as m'.

④ n

Specifies the number of overlap times in the field hold overlap writing mode or the field shifting overlap writing mode, or specifies the number of fields to be combined along either the one X axis or the one Y axis in the field combine writing mode. In the case of n', it specifies the same as n.

 If specifications of m and n are omitted, no special writing mode is applied.

4.3.44 SPPRM

(Optional; inapplicable in JBX-6000FS)

Format:	
SPPRM	[m1],[m2],[w],[p][,[sc][,[ss][,sfs]]]
	① ② ③ ④ ⑤ ⑥ ⑦
Examples:	
P(1)	'jeol.j5x',MULTI,3 (1)
SPPRM	2.0,4.0,0.24,0.12,,2 (2)

● Description of the example

- (1) The preceding P command specifies the file jeol.j5x and specifies to write the chip pattern contained in that file using the triple field shift overlap writing.
- (2) For the triple field shift overlap writing, the SPPRM command specifies the following conditions: maximum beam size = 2.0 μm , maximum beam area = 4.0 μm^2 , trapezoid writing slit width = 0.24 μm , trapezoid writing pitch = 0.12 μm , writing magnification = 1.0, shot-shift amount for the field shift overlap writing = half of the beam size, subfield-shift amount for the subfield-shift writing = half of the subfield size.

● Description of the command

The SPPRM command specifies the shot-partition parameters* for each writing chip pattern data. The SPPRM command specified here overrides the SPPRM command specified in the jobdeck if both concern the pattern data of the same pseudo pattern number. If the SPPRM command is specified neither in the jobdeck file nor in the schedule file for the pattern data of the pseudo pattern number, the default parameter values that are provided by the system will be used.

① m1

For the variable shaped e-beam machine, this parameter is used to specify the maximum value of the longer side of the shaped rectangle beam, for writing a chip. For the spot e-beam machine, this parameter is used to specify the maximum value of the side of the square beam scanning area that can be written by a single positioning deflection.

* In the case of the variable shaped e-beam machine, the parameters related to beam shaping are specified. In the case of the spot e-beam machine, the parameters related to beam scanning are those that can be written by a single positioning deflection.

- ② **m2****
 For the variable shaped e-beam machine, this parameter is used to specify the maximum value of the shaped rectangle beam area for writing a chip.
 In the case of the spot e-beam machine, this parameter is automatically set to the square of m1.
- ③ **w**
 For the variable shaped e-beam machine, this parameter is used to specify the length of the shorter side of the shaped rectangle beam for writing a trapezoid pattern.
 For the spot e-beam machine, specification of this parameter is inapplicable.
- ④ **p**
 For the variable shaped e-beam machine, this parameter is used to specify the shift pitch of the shaped rectangle beam in the shorter side direction for writing a trapezoid pattern.
 For the spot e-beam machine, specification of this parameter is inapplicable.
- ⑤ **sc**
 Specifies the magnification or reduction ratio of scaling for pattern writing.
 If you omit sc, the magnification or reduction ratios of scaling in the directions of both X and Y axes are set to 1.0.
 For the JBX-3040MV and later models, the product of the writing enlargement/reduction ratio specified in the JEOL52 (V3.1) format data and the value of this parameter is the scaling enlargement/reduction ratio to use for writing.
- ⑥ **ss**
 Specifies the shift amount for partition of the shaped rectangle beam or that for the spot beam scanning area when shot shift writing is performed. The value will be the reciprocal of the value obtained from the maximum beam area (m2) that corresponds to the pattern shape while the size of the maximum beam area is set to be 1 both in the horizontal direction and in the vertical direction. If you omit this parameter, shot shift writing will not be performed.
 If neither field fold overlap writing nor field shift overlap writing is applied to the pattern data, shot shift writing will also not be applied to it.
- ⑦ **sfs**
 The parameter sfs specifies the subfield-shift amount for the subfield-shift writing. For the parameter value, specify the reciprocal of a fraction whose numerator is 1, assuming that the subfield size is 1. The subfield-shift writing is disabled by default. For the pattern data for which the overlap writing and the field shift overlap writing are disabled, the subfield-shift writing is also disabled.
- ☞ For details about the shot shift writing, see the separate instruction manual.
 - ☞ The subfield-shift writing is an optional feature for the JBX-3030MV and a standard feature for the JBX-3040MV and later models. For details, see the manual, “General Statement” or the manual, “Operation” for your lithography system.

— CAUTION —

- **Be especially careful when you specify the parameters ① to ⑦,**

** In the case of the spot e-beam machine, the system ignores the specification of the parameter m2 even if it is specified.

because they are closely related to the writing precision.

4.3.45 STGPRM

(Optional/available only in JBX-3050MV and later models)

Format:

STGPRM [a1] [, [a2] [, [a3] [, [a4], a5]]]
① ② ③ ④ ⑤

Example:

P(1) 'jeol.j5x',MULTI/FP,2
STGPRM XRASTL,XRASTL

● Description of the example

- (1) The preceding P command specifies a double field shift overlap writing in which the full pass writing applies to the chip pattern contained in the jeol.j5x file.
- (2) In that writing, the substrate stage moves in the X raster sequence from the top left of the pattern to write the pattern of the first overlap, and the substrate stage moves in the X raster sequence from the top left of the pattern to write the pattern of the second overlap.

● Description of the command

The STGPRM command specifies the method of moving the substrate stage in the full pass writing. If the preceding P command specifies the full pass writing, the STGPRM command specifies how to move the substrate stage to write the pattern in the normal writing and the nth writing in the field shift overlap writing. If the full pass writing is not specified, only the a1 parameter takes effect; the writing is performed, with no distinction between the first and second overlaps, using the specified movement method (with XZIGTL specified, the writing is performed in the usual manner).

If the P command specifies the full pass writing and if the STGPRM command is omitted, the default values specified as system parameters are used. If the P command does not specify the full pass writing and if the STGPRM command is omitted, the writing is performed in the usual manner.

① a1

Here you can specify the direction in which to move the stage for the first overlap. You can use the following parameters to determine the movement method.

Parameter	Description
XRASTL	X raster writing from the top left of the pattern
XRASTR	X raster writing from the top right of the pattern
XRASBL	X raster writing from the bottom left of the pattern
XRASBR	X raster writing from the bottom right of the pattern
XZIGTL	X zigzag writing from the top left of the pattern
XZIGTR	X zigzag writing from the top right of the pattern
XZIGBL	X zigzag writing from the bottom left of the pattern

Parameter	Description
XZIGBR	X zigzag writing from the bottom right of the pattern

You can omit this parameter; in that case, the default value specified as a system parameter is used.

② a2

Here you can specify the direction in which to move the stage for the second overlap. The parameters are the same as with a1 (①). If the preceding P command does not specify the full pass writing, this parameter is ignored.

③ a3

Here you can specify the direction in which to move the stage for the third overlap. The parameters are the same as with a1 (①). If the preceding P command does not specify the full pass writing, this parameter is ignored.

④ a4

Here you can specify the direction in which to move the stage for the fourth overlap. The parameters are the same as with a1 (①). If the preceding P command does not specify the full pass writing, this parameter is ignored.

⑤ a5

Here you can specify the direction in which to move the stage for the fifth overlap. The parameters are the same as with a1 (①). If the preceding P command does not specify the full pass writing, this parameter is ignored.

 Omitting parameters

With both /FP and STGPRM omitted

⇒The conventional writing is performed.

With /FP omitted and STGPRM specified

⇒Only the first parameter of STGPRM takes effect, and using the movement method of that parameter, the writing is performed with no distinction between the first and second overlaps.

With /FP specified and STGPRM omitted

⇒The writing is performed using the default STGPRM value specified as a system parameter.

 Handling the SMASH writing

For a path for which SMASH is specified, the parameter that determines the movement method must be the same for all the patterns in the path. Any specification that is different will cause an error in the data-preparation stage before writing.

4.3.46 C

(Optional)

Format:		
C (i)	's'	
①	②	
Example:		
C (1)	'JEOL'	

- **Description of the example**

The character string “JEOL” will be written on the substrate at the position and with the attribute that are designated by character string number 1 of the CHAR command (CHAR/1).

- **Description of the command**

The C command specifies not only the position and attribute(s) of a character string by giving a character number, but also the character string to be written on the substrate.

The C command specified here will override the C command given in the common block and the layer block in the jobdeck file.

① i

Specifies the character string number that defines the position and attribute(s) of the character string.

② s

Specifies the contents of a character string that are to be written on the substrate.



The position and attribute(s) of the character string to be written are specified in the jobdeck. Refer to the CHAR command in the manual “Specifications for Jobdeck File Format”.

4.3.47 RESIST

(Optional)

Format:		
RESIST	s1 [,s2] [, a]	
	① ② ③	
Examples:		
RESIST	200.0,100.0	(1)
RESIST	2.4	(2)
RESIST	200.0,500.0,A	(3)
RESIST	200.0,2.4,L	(4)

- **Descriptions of the examples**

(1) The writing uses a resist with an area sensitivity of 200.0 $\mu\text{C}/\text{cm}^2$ and a line sensitivity of 100.0 nC/cm.

- (2) The writing uses a resist with an area sensitivity of $2.4 \mu\text{C}/\text{cm}^2$.
- (3) For writing, the JBX-6300 series and the JBX-5500 series uses a resist with an area sensitivity of $200.0 \mu\text{C}/\text{cm}^2$ and a line sensitivity of $500.0 \mu\text{C}/\text{cm}^2$.
- (4) For writing, the JBX-6300 series and the JBX-5500 series uses a resist with an area sensitivity of $200.0 \mu\text{C}/\text{cm}^2$ and a line sensitivity of $2.4 \text{ nC}/\text{cm}$.

● Description of the command

The RESIST command specifies the sensitivity of the resist to be used in pattern writing. The RESIST command specified here overrides the RESIST command in the jobdeck layer block.

① s1

Specifies the sensitivity of the resist for writing patterns with areas, such as rectangles and trapezoids.

② s2

Specifies the sensitivity of the resist for writing patterns without an area, such as single lines.

 The JBX-6000FS, the JBX-6300 series and the JBX-5500 series can write single lines.

 In the JBX-6000FS, the line sensitivity of the resist is represented by nC/cm . In the JBX-6000FS, the JBX-6300 series and the JBX-5500 series, s2 must not be omitted. In the JBX-9000MV and the JBX-9300FS, even if s2 is specified, it will not be applied.

③ a (available in the JBX-6300 series and JBX-5500 series)

Specifies the unit of resist line sensitivity. By default, A ($\mu\text{C}/\text{cm}^2$) is specified.

Unit of pattern with no area	$\mu\text{C}/\text{cm}^2$	nC/cm
Statement	A	L

 If you set the SHOT command to automatic mode to determine the writing time per electron-beam shot (shot time) by the resist sensitivity and the electron-beam current, you must give the RESIST command either in the jobdeck layer block or schedule cassette block.

4.3.48 SHOT

(Optional)

Format:			
SHOT	{ A [, s] S, t1, t2 [, s] V, r, x, y }		
	① ② ③④⑤ ⑥⑦⑧⑨		
Examples:			
SHOT	A, 2		(1)
SHOT	S, 200, 500, 2		(2)
SHOT	V, 100, 100, 100		(3)

● Descriptions of the examples

- (1) The writing time per e-beam shot (the shot time) is to be determined by the sensitivity of the resist and the e-beam current for writing, and the beam scanning step (shot pitch) 2 is to be used.
- (2) For the JBX-6000FS, the JBX-6300 series, the JBX-9300FS and the JBX-5500 series e-beam machine, the shot times for writing area patterns and single lines are to be 200 ns and 500 ns, respectively, and the beam scanning step 2 is to be used.
- (3) For the JBX-9000MV or the JBX-3030MV, the shot times for rectangles, X trapezoids, and Y trapezoids are to be 100 ns, 100 ns, and 100 ns.

● Description of the command

The SHOT command specifies the writing time per e-beam shot (the shot time) and the beam scanning step for the spot e-beam machine. There are two modes for specifying the shot times: one is called automatic mode in which the shot time is determined from the sensitivity of the resist and the e-beam current for writing, and the other is called manual mode in which the shot time can manually be specified according to the shape of the pattern. When this command is omitted, the automatic mode will be used. The shot time specified here overrides that specified by the SHOT command of the jobdeck layer block.

① A

Specifies that the shot time is to be determined from the sensitivity of the resist and the amount of beam current for writing.

② s

Specifies the beam scanning step (shot pitch) for pattern writing with the spot e-beam machine. When this parameter is omitted, the scanning step is set to 1. This specification is ignored in the variable shaped e-beam machine.

 If the parameter s is omitted when using JBX-6000FS, the beam scanning interval specified using the PREAD program will be applied to individual chip patterns to be written.

③ S

Specifies that the shot time is to be applied to the spot e-beam machine.

④ t1

Specifies the shot time for area patterns used to be in the spot e-beam machine.

⑤ t2

Specifies the shot time for single line patterns to be used in the spot e-beam machine.

 In the case of JBX-6000FS, JBX-6300 series and JBX-5500 series the parameter t2 must be specified when you use the SHOT command; however, in the case of JBX-9300FS, this parameter is not used even if it is specified.

⑥ V

Specifies that the shot time is to be applied to the variable shaped e-beam machine.

⑦ r

Specifies the shot time for rectangular patterns to be used in the variable shaped e-beam machine.

⑧ x

Specifies the shot time for X trapezoidal patterns to be used in the variable shaped e-beam machine.

⑨ y

Specifies the shot time for Y trapezoidal patterns to be used in the variable shaped e-beam machine.

 Other than SHOT (the basic shot time) of this section, three shot time correction commands, namely MODULAT (shot time modulation according to shot rank), SHTCOR (shot time correction according to pattern shape), and SHTOFS (shot time offset) are provided.

 For information about these commands, refer to Sects. 4.3.45, 4.3.46, and 4.3.14, respectively.

4.3.49 MODULAT

(Optional)

Format:

```
t:  MODULAT  {((r,v)[,⋯]) |
①              ②③
–              (((r0,r1,r2),(v0,v1,v2))[,⋯])}
                ④⑤⑥ ⑦⑧⑨
```

or

```
't': MODULAT  {((r,v)[,⋯]) |
–              (((r0,r1,r2),(v0,v1,v2))[,⋯])}
```

Examples:

```
MDLT01:  MODULAT  ((0,-10)) (1)
MDLT02:  MODULAT  (((1,5,9),(-30,10,45)) (2)
–          ,((10,15,20),(-95,-50,3100)))
```

● Descriptions of the examples

- (1) Shot time modulation of -10% is added to the shot time of the pattern having shot rank number 0.
- (2) Shot time modulation is allotted to shot rank numbers 1 to 9, increasing geometrically by 10%, with the shot time modulation at shot rank 5 being the reference point (0%), and to the shot rank numbers 10 to 20, also increasing geometrically (but by -50%) with the shot time modulation at shot rank 15 being the reference point. Shot time modulation thus allotted is added to patterns having a corresponding shot rank number.

● Description of the command

The MODULAT command specifies modulation of shot time (defined by the SHOT command) for each shot rank of chip pattern data. Shot time modulation may be allotted individually to each shot rank number, or to shot rank number in a specific range at a certain fixed ratio.

Correspondence between chip pattern writing with shot time modulation and the MODULAT command is made by allotting a shot time modulation table name defined by the ASSIGN command in the common block of the jobdeck file.

The shot time modulation of 0% will be allotted for shot rank numbers that are not specified by this command.

Shot time for e-beam writing is obtained from the specified shot time (St) and the shot time modulation amount (V) as follows:

$$(\text{Shot time for writing}) = \text{St} \times \frac{100 + V}{100}$$

If the MODULAT command having the same shot time modulation table is also specified in a layer block of the jobdeck file, the specification issued here overrides that in the layer block of the jobdeck file. However, shot rank modulation specified in the jobdeck file will be allotted to the shot rank numbers that are not defined by the MODULAT command in the cassette block of the schedule file. Even though you specify the name of a shot time modulation table here, if that is not specified by the ASSIGN command in the jobdeck file, it will be invalid.

- ① t
Specifies the name of a shot time modulation table. If the character string that indicates the shot time modulation table name begins with a character other than a letter, enclose the entire character string in single quotes (')*.
- ② r
Specifies a shot rank number.
- ③ v
Specifies a shot time modulation amount.
- ④ r0
Specifies the starting number of a series of shot rank numbers.
- ⑤ r1
Specifies the shot rank number where the shot time modulation to be added is 0.
- ⑥ r2
Specifies the last number of the shot rank numbers.
- ⑦ v0
Specifies the lower limit of the shot time modulation amounts.
- ⑧ v1
Specifies the percent change (ratio) of shot time modulation.
- ⑨ v2
Specifies the upper limit of the shot time modulation amount.

 By specifying the parameters r0 to r2, and v0 to v2, the shot time modulation (V) for an arbitrary shot rank number (R) is determined from the following equation:

$$V = \left\{ \left(1 + \frac{v1}{100} \right)^{(R-r1)} - 1 \right\} \times 100$$

The parameters r0 and r2, and v0 and v2 are limited as follows:

$$r0 \leq r2, \text{ and } v0 \leq v2$$

* This description method is available only in the JBX-3030MV.

4.3.50 MMDOFS

(Optional; applicable only in JBX-9300FS)

Format:	
MMDOFS	{P(i) [(x,y)] GLM [P=(px,py), Q=(qx,qy)]
	①② ③④ ⑤ ⑥⑦ ⑧⑨
–	[(R=(rx,ry), S=(sx,sy))] CHM(n) [(cx,cy)] }
	⑩⑪ ⑫⑬ ⑭⑮ ⑯⑰
Examples:	
MMDOFS	P(1),(-1000,1000) (1)
MMDOFS	GLM,P=(-1000,0),Q=(1000,0) (2)
MMDOFS	CHM(2),(1000,-1000) (3)

● Descriptions of the examples

- (1) Mix-and-match distortion correction is to be carried out when writing the chip pattern assigned the pseudo pattern number 1 in the specified jobdeck. In that process, the mix-and-match distortion correction amounts (hereinafter called the correction amounts) will be applied to the chip pattern writing in the condition that the center of the chip pattern is located at X = -1000 μm and Y = 1000 μm in the relative coordinate values (X and Y coordinate values increase toward the right and top, respectively) from the center of the mix-and-match distortion correction map (hereinafter called the correction map).
- (2) Mix-and-match distortion correction is to be carried out when detecting the global marks P and Q specified in the jobdeck. In that process, the correction amounts will be applied to global mark P detection in the condition that the center of the global mark P is located at X = -1000 μm and Y = 0 μm in the relative coordinate values (X and Y coordinate values increase toward the right and top, respectively) from the center of the correction map. Also, the correction amounts will be applied to global mark Q detection in the condition that the center of the global mark Q is located at X = 1000 μm and Y = 0 μm in the relative coordinate values (X and Y coordinate values increase toward the right and top, respectively) from the center of the correction map.
- (3) Mix-and-match distortion correction is to be carried out when detecting the chip marks defined by the chip mark definition (specified by the CHMPOS command) that is the second one from the top of the line in the specified jobdeck. In that process, the correction amount will be applied to chip mark detection in the condition that the coordinate origin at which the each chip mark is defined (the array point in the chip array definition) is located at X = 1000 μm and Y = -1000 μm in the relative coordinate values (X and Y coordinate values increase toward the right and top, respectively) from the center of the correction map.

● Description of the command

The MMDOFS command specifies the chip patterns and the various marks to be written using the mix-and-match distortion correction function, and the correction amounts that are to be individually applied.

When this command is omitted, the correction amounts will be applied in the condition

that the center of the correction map coincides with the center of the chip patterns to be written and with the center of various marks to be used.

- ① **P**
Specifies the mix-and-match distortion correction amount for chip pattern writing.
- ② **i**
Specifies the pseudo-pattern number in the specified jobdeck for assigning the chip patterns that are to be written using the mix-and-match distortion correction function.
- ③ **x**
Specifies which position's correction amount in the correction map is to be applied to the chip pattern so as to specify the correction amount to be applied to the chip pattern to be written. The position is to be specified on the X axis on the correction map at which the chip pattern center exists using the relative X coordinate value (the value increases toward the right) with its origin being the center of the correction map. When this parameter is omitted, the value will be set to 0.
- ④ **y**
Specifies which position's correction amount in the correction map is to be applied to the chip pattern so as to specify the correction amount to be applied to the chip pattern to be written. The position is to be specified on the Y axis on the correction map at which the chip pattern center exists using the relative Y coordinate value (the value increases toward the top) with its origin being the center of the correction map. When this parameter is omitted, the value will be set to 0.
- ⑤ **GLM**
Specifies the mix-and-match distortion correction amount for detecting the global mark.
- ⑥ **px**
Specifies which position's correction amount in the correction map is to be applied to detect the global mark P so as to specify the correction amount to be applied to the global mark P to be detected. The position is to be specified on the X axis in the correction map at which the global mark P center exists using the relative X coordinate value (the value increases toward the right) with its origin being the center of the correction map. When this parameter is omitted, The value will be set to 0.
- ⑦ **py**
Specifies which position's correction amount in the correction map is to be applied to detect the global mark P so as to specify the correction amount to be applied to the global mark P to be detected. The position is to be specified on the Y axis in the correction map at which the global mark P center exists using the relative Y coordinate value (the value increases toward the top) with its origin being the center of the correction map. When this parameter is omitted, The value will be set to 0.
- ⑧ **qx**
Specifies which position's correction amount in the correction map is to be applied to detect the global mark Q so as to specify the correction amount to be applied to the global mark P to be detected. The position is to be specified on the X axis in the correction map at which the global mark Q center exists using the relative X coordinate value (the value increases toward the right) with its origin being the center of the correction map. When this parameter is omitted, The value will be set to 0.

- ⑨ **qy**
Specifies which position's correction amount in the correction map is to be applied to detect the global mark Q so as to specify the correction amount to be applied to the global mark P to be detected. The position is to be specified on the Y axis in the correction map at which the global mark Q center exists using the relative Y coordinate value (the value increases toward the top) with its origin being the center of the correction map. When this parameter is omitted, The value will be set to 0.
- ⑩ **rx**
Specifies which position's correction amount in the correction map is to be applied to detect the global mark R so as to specify the correction amount to be applied to the global mark P to be detected. The position is to be specified on the X axis in the correction map at which the global mark R center exists using the relative X coordinate value (the value increases toward the right) with its origin being the center of the correction map. When this parameter is omitted, The value will be set to 0.
- ⑪ **ry**
Specifies which position's correction amount in the correction map is to be applied to detect the global mark R so as to specify the correction amount to be applied to the global mark P to be detected. The position is to be specified on the Y axis in the correction map at which the global mark R center exists using the relative Y coordinate value (the value increases toward the top) with its origin being the center of the correction map. When this parameter is omitted, The value will be set to 0.
- ⑫ **sx**
Specifies which position's correction amount in the correction map is to be applied to detect the global mark S so as to specify the correction amount to be applied to the global mark P to be detected. The position is to be specified on the X axis in the correction map at which the global mark S center exists using the relative X coordinate value (the value increases toward the right) with its origin being the center of the correction map. When this parameter is omitted, The value will be set to 0.
- ⑬ **sy**
Specifies which position's correction amount in the correction map is to be applied to detect the global mark S so as to specify the correction amount to be applied to the global mark P to be detected. The position is to be specified on the Y axis in the correction map at which the global mark S center exists using the relative Y coordinate value (the value increases toward the top) with its origin being the center of the correction map. When this parameter is omitted, The value will be set to 0.
- ⑭ **CHM**
Specifies the mix-and-match distortion correction amount for detecting the chip mark.
- ⑮ **n**
Specifies the chip mark to be detected by using the mix-and-match distortion correction amount for detecting chip pattern writing. The chip mark is to be sequentially specified in the order of appearance of the applicable chip mark definitions (by the CHMPOS command) issued in the jobdeck file.
- ⑯ **cx**
Specifies which position's correction amount in the correction map is to be applied to chip mark detection so as to specify the correction amount to be applied to each

chip mark to be detected. The position is to be specified on the X axis on the correction map at which the coordinate origin that is the array point of each chip array definition and defines each chip mark position exists using the relative X coordinate value (the value increases toward the right) with its origin being the center of the correction map. When this parameter is omitted, the value will be set to 0.

⑰ cy

Specifies which position's correction amount in the correction map is to be applied to chip mark detection so as to specify the correction amount to be applied to each chip mark to be detected. The position is to be specified on the Y axis on the correction map at which the coordinate origin that is the array point of each chip array definition and defines each chip mark position exists using the relative Y coordinate value (the value increases toward the top) with its origin being the center of the correction map. When this parameter is omitted, the value will be set to 0.

4.3.51 DEFOCUS

(Optional)

Format:

DEFOCUS d
 ①

Example:

DEFOCUS -1000

- **Description of the example**

The focus value of the electron optical system of the e-beam lithography system is reduced from the present setting by 1000 points.

- **Description of the command**

The DEFOCUS command specifies a value (the defocus value) that is to be added to or subtracted from the focus value of the electron optical system of the e-beam lithography system. To carry out pattern writing using the GHOST™* method, a technique to correct for the proximity effect, the focus setting is deliberately changed. If this command is not given, the defocus value 0 will be set.

When the DEFOCUS command is specified both in the layer block of the jobdeck file and in the cassette block of the schedule file, the value specified in the cassette block of the schedule file overrides that specified in the layer block of the jobdeck file.

① d

Specifies the amount of adjustment to focus setting to be added or subtracted.

* GHOST™ is a registered trademark of Hewlett-Packard Company.

4.4 FOOTER COMMAND

4.4.1 END

(Required)

Format:	
END	① [n]
Example:	
END	2

- **Description of the example**

This statement terminates the writing control program (EXPOSE) after pattern writing of all the specified cassettes in the magazine is done and the substrate stored on shelf number 2 of the autoloader (ALD) is loaded on the stage.

- **Description of the command**

The END command terminates the schedule and also specifies a substrate that is to be loaded on the stage of the work chamber after completion of pattern writing. To specify the substrate, designate the shelf number of the stocker of the substrate- handling system or the shelf number of the autoloader (ALD) of the e-beam lithography system, according to the designation by the MTRL header command.

This command must be given at the end of a series of schedule commands and cannot be omitted. All records that follow the END command line do not affect the schedule.

① n

Specifies the shelf number of the stocker of the substrate-handling system or the shelf number of the autoloader of your e-beam lithography system in which the substrate to be set on the stage of the work chamber after completion of pattern writing is stored. When this parameter is omitted, no substrate is to be loaded on the stage after completion pattern writing.

 If your e-beam lithography system does not have an autoloader, the designation of the shelf number n is ignored.

When you wish to write substrates that are continuously supplied through the inline system (that is, when you have specified ILN in the MTRL command line), the designation of the parameter n is inapplicable. The result will be the same as when the shelf number n is omitted.

4.5 CONTROL CHARACTERS

4.5.1 – (hyphen)

(Optional)

Format:			
–		Parameters (which are continued)	
Examples:			
MDLT01:	MODULAT	((0,–10),(1,–20),	(1)
–		(2,–30),(3,–40))	(2)

- **Description of the example**

When the parameter list of a command is too long to write in one physical record, the parameter list can be divided into the initial line (1) and the continuation line (2) as shown above. In the above, the label name “MDLT01” with the MODULAT command is used as an example, but the parameter lists of other label names and commands can be continued using the hyphen as well.

- **Description of command**

The hyphen (–) placed in the first column of a line indicates that the character string that follows this hyphen is continued from the immediately previous line. Only the parameter list can be continued using this code. This hyphen must always be placed in the first column of the line.

4.5.2 ; (semicolon)

(Optional)

Format:			
		; An arbitrary comment	
Example:			
P (1)	'dummy.j5x'	; FOR DUMMY	

- **Description of the example**

The comment, “FOR DUMMY,” is added to the P command after the semicolon (;).

- **Description of the command**

The text from the semicolon (;) to the end of the line (record) is treated as a comment. The comment does not affect the schedule file.

4.5.3 @ (at mark)

(Optional)

Format:

@ 'f'

①

Example:

@ 'include.inc'

- **Description of the example**

The file include.inc is inserted in the place of the record marked by the at mark.

- **Description of the command**

An at mark in a record (line) designates that the specific file named by this statement is to be inserted in the place of the record. The at mark must always be placed in the first column of the record and the schedule must not have any contradiction after the insertion of the file.

① f

The name of the file to be inserted.

5 APPENDICES

5.1 Limits of Commands

■ Maximum number of commands that can be repeated

**Table 6 Maximum number of commands that can be repeated
(Models except JBX-3030MV and JBX-3040MV)**

Items	Command	Limit
CALIB menu name	PATH	Up to 32/cassette
Writing pattern definition	P	Up to 99/cassette
Writing character string definition	C	Up to 4/ cassette
Shot modulation table	MODULAT	Up to 1000/cassette

**Table 7 Maximum number of commands that can be repeated
(JBX-3030MV and JBX3040MV)**

Items	Command	Limit
CALIB menu name	PATH	Up to 2000*/cassette
Writing pattern definition	P	Up to 2000**/cassette
Writing character string definition	C	Up to 4/ cassette
Shot modulation table	MODULAT	Up to 2000/cassette

■ Multiple occurrences of an identical command

In each cassette block, the last specified one has higher priority than the others.

* For the JBX-3030MV, the maximum might be limited to 256 depending on the hardware configuration. For the JBX-3040MV and later models, the maximum is 2000.

** For the JBX-3030MV, the maximum might be limited to 150 or 300 depending on the hardware configuration. For the JBX-3040MV and later models, the maximum is 2000.

5.2 Limits of Parameters

5.2.1 For JBX-6000FS

Terminology	
min:	minimum value
max:	maximum value
step	size
unit	unit
slct	selections
fix	fixed value

Table 8 Limits of parameters for JBX-6000FS

Command	Parameter	Limits	Remarks (JBX-6000FS)
MAGAZIN	m	Up to 9 uppercase letters, numbers, and symbols; must start with a letter.	You cannot specify a character string that includes a single quote (').
MTRL	m	Inapplicable	
#	n	min = 1 max = 12 step = 1	*
	m	Same as n	Same as n
%	s	slct = 2, 3, 4, 5, or 6	
	p	An uppercase letter	A to J (10 kinds)
LYR	l	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
JDF	j	Same as LYR l	Same as LYR 1
	n	min = 1 max = 100 step = 1	
EOS	m	min = 1 max = 8 step = 1	
	t	min = 0 max = 9 step = 1	
	c	Inapplicable	
ACC	v	Inapplicable	
CALPRM	c	Inapplicable	
DEFMODE	m	Inapplicable	

* The limit values depend on the type of the attached autoloader. The values mentioned here are those for JBX-6000FS.

Command	Parameter	Limits	Remarks (JBX-6000FS)
RESTOR	m	slct = 0 or 1	
	sw	Inapplicable	
	a	Inapplicable	
OFFSET	dx	min = -104900.000 max = 76200.000 step = 0.001 unit = μm	*
	dy	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	*
MCOR	gx	min = -100.000 max = 100.000 step = 0.001 unit = $\mu\text{m}/100\text{ mm}$	
	gy	Same as gx	
	ry	Same as gx	
HEIGHT	f	min = -4095 max = 4095 step = 1	
	gx	Same as f	
	gy	Same as f	
	rx	Same as f	
	ry	Same as f	
	sx	Inapplicable	
	sy	Inapplicable	
SUBHEI	gx	Inapplicable	
	gy	Inapplicable	
	rx	Inapplicable	
	ry	Inapplicable	
	shx	Inapplicable	
	shy	Inapplicable	
SHTCOR	l	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	a	Same as l	
	r	Inapplicable	
	tx	Inapplicable	
	ty	Inapplicable	

* The limit values depend on the workstage size of the e-beam lithography machine. The values mentioned here are those for JBX-6000FS.

Command	Parameter	Limits	Remarks (JBX-6000FS)
STEPPER	s	Inapplicable	
MMDPRM	m	Inapplicable	
SPCPRM	spc	Inapplicable	
SHTOFS	st	Inapplicable	
PECPRM	pec	Inapplicable	
FOGPRM	fog	Inapplicable	
PRCPRM	prc	Inapplicable	
PRJPRM	prj	Inapplicable	
LECPRM	lec	Inapplicable	
GLECPRM	gl	Inapplicable	
ANG2SIZ	a2s	Inapplicable	
SSCPRM	scc	Inapplicable	
PEDPRM	ped	Inapplicable	
WARMUP	t	min = 0.1 max = 3276.7 step = 0.1 unit = minutes	
	m	min = 0 max = 2 step = 1	Even if you set m = 0 or 2, m is automatically set to 1.
LBC	sw	slct = ON or HOLD	
HSWITCH	swg	Inapplicable	
	swc	Inapplicable	
GLMPRT	ps	min = 0.010 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	fw	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	m	slct = 0 or 1	
PLMPRT	ps	Inapplicable	
	fw	Inapplicable	
	m	Inapplicable	
CHMPRT	ps	min = 0.010 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	

Command	Parameter	Limits	Remarks (JBX-6000FS)
	fw	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	m	slct = 0 or 1	
GLMDET	sw	slct = A, S, M, or C	
PLMDET	sw	Inapplicable	
CHMDET	sw	fix = A	
	e	fix = 0 or 1	
	n	min = 1 max = 9 step = 1	
WFR3M	m	min = 1 max = 9 step = 1	
CHIPAL	m	Inapplicable	
CDRIFT	a1	min = 0.1 max = 99.9 step = 0.1 unit = %	
	a2	Same as a1	
	m	slct = A or S	
PDRIFT	a1	min = 0.001 max = 99.999 step = 0.001 unit = μm	
	a2	Same as a1	
	m	slct = A or S	
TDRIFT	a1	min = 0.01 max = 99.99 step = 0.01 unit = $^{\circ}\text{C}$	
	a2	Same as a1	
	m	slct = A or S	
PATH	n	min = 1 max = 32 step = 1	
	m	Up to 6 uppercase letters, numbers, and symbols.	Character strings that are reserved as command names cannot be used.
P	i	min = 1 max = 99 step = 1	
	p	Up to 24 uppercase letters, numbers, and symbols acceptable as a pattern ID name.	Single quote (') cannot be included in an ID name.

Command	Parameter	Limits	Remarks (JBX-6000FS)
SPPRM	m	Inapplicable	
	n	Inapplicable	
	m1	Inapplicable	
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	Inapplicable	
	ss	Inapplicable	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	*
RESIST	s1	min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	s2	min = 0.001 max = 1000.000 step = 0.001 unit = nC/cm	
	a	Inapplicable	
SHOT	s	min = 1 max = 8 step = 1	**
	t1	min = 167 max = 3900000 step = 1 unit = ns	
	t2	Same as t1	
	r	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	

* The character size is 1.6 mm × 1.6 mm, and the writing pitch is 1.6 mm.

** In JBX-5DII (2 MHz), max = 40; in both JBX-5DII (6 MHz) and JBX-5000LS (6 MHz), max = 20; in both JBX-5FE (6 MHz) and JBX-6000FS (12 MHz), max = 8.

Command	Parameter	Limits	Remarks (JBX-6000FS)
MODULAT	t	Up to 6 uppercase letters, numbers, and symbols.	Character strings that are reserved as command names cannot be used.
	r	min = 0 max = 63 step = 1	In single line writing, figures greater than 16 are ignored.
	v	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r0	min = 0 max = r2 step = 1	In single line writing, figures greater than 16 are ignored.
	r1	min = 0 max = 63 step = 1	In single line writing, figures greater than 16 are ignored.
	r2	min = r0 max = 63 step = 1	In single line writing, figures greater than 16 are ignored.
	v0	min = -99.9 max = v2 step = 0.1 unit = %	
	v1	min = -50.0 max = 100.0 step = 0.1 unit = %	
	v2	min = v0 max = 3200.0 step = 0.1 unit = %	
MMDOFS	i	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
	px	Inapplicable	
	py	Inapplicable	
	qx	Inapplicable	
	qy	Inapplicable	
	rx	Inapplicable	
	ry	Inapplicable	
	sx	Inapplicable	
	sy	Inapplicable	
	n	Inapplicable	
	cx	Inapplicable	

Command	Parameter	Limits	Remarks (JBX-6000FS)
	cy	Inapplicable	
DEFOCUS	d	min = -4095 max = 4095 step = 1	Inapplicable at present; designation is ignored.
@	f	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
END	n	min = 1 max = 12 step = 1	*

* The limit values depend on the type of the attached autoloader. The values mentioned here are those for JBX-6000FS.

5.2.2 For JBX-9000MV

Terminology	
min:	minimum value
max:	maximum value
step	size
unit	unit
slct	selections
fix	fixed value

Table 9 Limits of parameters for JBX-9000MV

Command	Parameter	Limits	Remarks (JBX-9000MV)
MAGAZIN	m	Up to 9 uppercase letters, numbers, and symbols; must begin with a letter.	You cannot specify a character string that includes a single quote (').
MTRL	m	Inapplicable	
#	n	min = 1 max = 10 step = 1	*
	m	Same as n	Same as n
%	s	slct = 2, 3, 4, 5 or 6	
	p	An uppercase letter	A to J (10 kinds)
LYR	l	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
JDF	j	Same as LYR l	Same as LYR 1
	n	min = 1 max = 100 step = 1	
EOS	m	Inapplicable	
	t	Inapplicable	
	c	Inapplicable	
ACC	v	Inapplicable	
CALPRM	c	A character string of up to 63 characters consisting of printable uppercase and lowercase letters, numbers, and symbols that exclude not only any non-pictorial character including spaces but also slash, backward slash, and single quotation.	
DEFMODE	m	Inapplicable	

* The devices that can be selected depend on the type of the attached substrate handling system. When a mask database for the management of the substrate is used, n is not valid.

Command	Parameter	Limits	Remarks (JBX-9000MV)
RESTOR	m	Inapplicable	
	sw	Inapplicable	
	a	Inapplicable	
OFFSET	dx	min = -140000.000 max = 120000.000 step = 0.001 unit = μm	
	dy	min = -120000.000 max = 120000.000 step = 0.001 unit = μm	
MCOR	gx	min = -100.000 max = 100.000 step = 0.001 unit = $\mu\text{m}/100\text{ mm}$	
	gy	Same as gx	
	ry	Same as gx	
HEIGHT	f	min = -4095 max = 4095 step = 1	
	gx	Same as f	
	gy	Same as f	
	rx	Same as f	
	ry	Same as f	
	sx	Inapplicable	
	sy	Inapplicable	
SUBHEI	gx	Inapplicable	
	gy	Inapplicable	
	rx	Inapplicable	
	ry	Inapplicable	
	shx	Inapplicable	
	shy	Inapplicable	
SHTCOR	l	Inapplicable	
	a	Inapplicable	
	r	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	tx	Same as r	
	ty	Same as r	

Command	Parameter	Limits	Remarks (JBX-9000MV)
STEPPER	s	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
MMDPRM	m	Inapplicable	
SHTOFS	st	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
SPCPRM	spc	Inapplicable	
PECPRM	pec	Same as SHTOFS st	Same as SHTOFS st
FOGPRM	fog	Same as SHTOFS st	Same as SHTOFS st
PRCPRM	prc	Same as SHTOFS st	Same as SHTOFS st
PRJPRM	prj	Same as SHTOFS st	Same as SHTOFS st
LECPRM	lec	Inapplicable	
GLECPRM	gl	Inapplicable	
ANG2SIZ	a2s	Inapplicable	
SSCPRM	scc	Inapplicable	
PEDPRM	ped	Inapplicable	
WARMUP	t	min = 0.1 max = 3276.7 step = 0.1 unit = minutes	
	m	min = 0 max = 2 step = 1	
LBC	sw	Inapplicable	
HSWITCH	swg	Inapplicable	
	swc	Inapplicable	
GLMPRT	ps	min = 0.010 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	fw	min = 0.000 max = 750.000 step = 0.001 unit = μm	
	m	slct = 0 or 1	
PLMPRT	ps	min = 0.010 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	

Command	Parameter	Limits	Remarks (JBX-9000MV)
	fw	min = 0.000 max = 750.000 step = 0.001 unit = μm	
	m	slct = 0 or 1	
CHMPRT	ps	min = 0.100 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	fw	min = 0.000 max = 750.000 step = 0.001 unit = μm	
	m	slct = 0 or 1	
GLMDET	sw	slct = A, S, M, or C	
PLMDET	sw	fix = A	
CHMDET	sw	fix = A	
	e	slct = 0 or 1	
	n	min = 1 max = 9 step = 1	
WFR3M	m	Inapplicable	
CHIPAL	m	slct = 0, 1, 4, S, V1, or V4	
CDRIFT	a1	min = 0.1 max = 99.9 step = 0.1 unit = %	
	a2	Same as a1	
	m	slct = A or S	
PDRIFT	a1	min = 0.001 max = 99.999 step = 0.001 unit = μm	
	a2	Same as a1	
	m	slct = A or S	
TDRIFT	a1	min = 0.01 max = 99.99 step = 0.01 unit = $^{\circ}\text{C}$	
	a2	Same as a1	
	m	slct = A or S	
PATH	n	min = 1 max = 32 step = 1	

Command	Parameter	Limits	Remarks (JBX-9000MV)
	m	Up to 6 uppercase letters, numbers, and symbols.	Character strings that are reserved as command names cannot be used.
P	i	min = 1 max = 99 step = 1	
	p	A character string can consist of up to 24 uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
	m	slct = MULTI, GATHER, or OVERLAP	
	n	min = 2 max = 10 step = 1	If m = MULTI, max = 5 (as maximum). If m = OVERLAP, max = 8 (as maximum)
SPPRM	m1	min = 0.002 max = 4.000 step = 0.002 unit = μm	Where the accelerating voltages is 50 kV.*
	m2	min = 0.000004 max = 16.000000 step = 0.000004 unit = μm^2	Where the accelerating voltages is 50 kV.**
	w	min = 0.008 max = 2.040 step = 0.008 unit = μm	Where the accelerating voltages is 50 kV.*** The values should be multiples of p.
	p	min = 0.008 max = 2.040 step = 0.008 unit = μm	Where the accelerating voltages is 50 kV.***
	sc	min = 0.500000 max = 1.200000 step = 0.000001	
	ss	min = 1 max = 5 step = 1	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	

* If the accelerating voltage is 25 kV, min = 0.004, max = 4.000, step = 0.004.

** If the accelerating voltage is 25 kV, min = 0.000016, max = 16.000000, step = 0.000016.

*** If the accelerating voltage is 25 kV, min = 0.016, max = 4.080, step = 0.0016.

Command	Parameter	Limits	Remarks (JBX-9000MV)
C	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	****
RESIST	s1	min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	s2	Inapplicable	
SHOT	s	Inapplicable	
	t1	Inapplicable	
	t2	Inapplicable	
	r	min = 20 max = 16382000 step = 5 unit = ns	
	x	Same as r	
	y	Same as r	
MODULAT	t	Up to 6 uppercase letters, numbers, and symbols.	Character strings that are reserved as command names cannot be used.
	r	min = 0 max = 63 step = 1	
	v	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r0	min = 0 max = r2 step = 1	
	r1	min = 0 max = 63 step = 1	
	r2	min = r0 max = 63 step = 1	
	v0	min = -99.9 max = v2 step = 0.1 unit = %	
	v1	min = -50.0 max = 100.0 step = 0.1 unit = %	

**** The character size is 1.0 mm × 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-9000MV)
	v2	min = v0 max = 3200.0 step = 0.1 unit = %	
MMDOFS	i	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
	px	Inapplicable	
	py	Inapplicable	
	qx	Inapplicable	
	qy	Inapplicable	
	rx	Inapplicable	
	ry	Inapplicable	
	sx	Inapplicable	
	sy	Inapplicable	
	n	Inapplicable	
	cx	Inapplicable	
	cy	Inapplicable	
DEFOCUS	d	min = -4095 max = 4095 step = 1	Inapplicable at present. Designation is neglected.
@	f	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
END	n	min = 1 max = 10 step = 1	*

* The devices that can be selected depend on the type of the attached substrate handling system. When a mask database for the management of the substrate is used, n is not valid.

5.2.3 For JBX-9300FS

Terminology	
min:	minimum value
max:	maximum value
step	size
unit	unit
slct	selections
fix	fixed value

Table 10 Limits of parameters for JBX-9300FS

Command	Parameter	Limits	Remarks (JBX-9300FS)
MAGAZIN	m	Up to 9 uppercase letters, numbers, and symbols; must begin with a letter.	You cannot specify a character string that includes a single quote (').
MTRL	m	slct = STKR, CRR, ILN, or ALD	*
F#	n	min = 1 max = 26 step = 1	**
	m	min = 1 max = 999 step = 1	***
%	s	slct = 2, 3, 4, 5, or 6	
	p	An uppercase letter	A to J (10 kinds)
LYR	l	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
JDF	j	Same as LYR l	Same as LTR l
	n	min = 1 max = 99 step = 1	
EOS	m	Inapplicable	
	t	Inapplicable	
	c	Inapplicable	
ACC	v	slct = 25, 50, or 100 unit = kV	****

* The selectable device depends on the type of the attached substrate handling system.

** The maximum value depends on the parameter m of the MTRL command. If m is STKR, max = 4; if m is INL, max = 1; if m is ALD, max = 2.

*** The maximum value depends on the parameter m of the MTRL command. If m is STKR, max = 4; if m is CRR, max = 26; if m is ALD, max = 2.

**** If your JBX-9300FS does not have the 25 kV operation mode, which is an option, you cannot use that mode.

Command	Parameter	Limits	Remarks (JBX-9300FS)
CALPRM	c	A character string of up to 63 characters consisting of printable uppercase and lowercase letters, numbers, and symbols that exclude not only any non-pictorial character string including spaces, but also slashes, backward slashes, and single quotations.	
DEFMODE	m	slct = 1 or 2	
RESTOR	m	slct = 0 or 1	
	sw	slct = AL, CF or CS	
	a	min = 0.1 max = 20.0 step = 0.1 unit = %	
OFFSET	dx	min = -140000.000 max = 120000.000 step = 0.001 unit = μm	
	dy	min = -120000.000 max = 120000.000 step = 0.001 unit = μm	
MCOR	gx	min = -100.000 max = 100.000 step = 0.001 unit = $\mu\text{m}/100\text{ mm}$	
	gy	Same as gx	
	ry	Same as gx	
HEIGHT	f	min = -65535 max = 65535 step = 1	
	gx	Same as f	
	gy	Same as f	
	rx	Same as f	
	ry	Same as f	
	sx	min = -1000000.0 max = 1000000.0 step = 0.1 unit = nm	
	sy	Same as sx	
SUBHEI	gx	min = -65535 max = 65535 step = 1	
	gy	Same as gx	
	rx	min = -16383 max = 16383 step = 1	

Command	Parameter	Limits	Remarks (JBX-9300FS)
	ry	Same as rx	
	shx	min = -4095 max = 4095 step = 1	
	shy	Same as shx	
SHTCOR	l	Inapplicable	
	a	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r	Inapplicable	
	tx	Inapplicable	
	ty	Inapplicable	
STEPPER	s	Inapplicable	
MMDPRM	m	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
SHTOFS	st	Same as MMDPRM m	Same as MMDPRM m
SPCPRM	spc	Inapplicable	
PECPRM	pec	Inapplicable	
FOGPRM	fog	Inapplicable	
PRCPRM	prc	Inapplicable	
PRJPRM	prj	Inapplicable	
LECPRM	lec	Inapplicable	
GLECPRM	gl	Inapplicable	
ANG2SIZ	a2s	Inapplicable	
SSCPRM	scc	Inapplicable	
PEDPRM	ped	Inapplicable	
WARMUP	t	min = 0.1 max = 3276.7 step = 0.1 unit = minutes	
	m	slct = 1 or 2	
LBC	sw	slct = ON or HOLD	
H SWITCH	swg	slct = ON or OFF	
	swc	Same as swg	
GLMPRT	ps	min = 0.010 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	

Command	Parameter	Limits	Remarks (JBX-9300FS)
	fw	min = 0.000 max = 750.000 step = 0.001 unit = μm	
	m	slct = 0 or 1	
PLMPRT	ps	Inapplicable	
	fw	Inapplicable	
	m	Inapplicable	
CHMPRT	ps	min = 0.010 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	fw	min = 0.000 max = 750.000 step = 0.001 unit = μm	
	m	slct = 0 or 1	
GLMDET	sw	slct = A, S, M, or C	
PLMDET	sw	Inapplicable	
CHMDET	sw	fix = A	
	e	slct = 0 or 1	
	n	min = 1 max = 999 step = 1	
WFR3M	m	Inapplicable	
CHIPAL	m	slct = 0, 1, 4, S, V1, or V4	
CDRIFT	a1	min = 0.1 max = 99.9 step = 0.1 unit = %	
	a2	Same as a1	
	m	slct = A or S	
PDRIFT	a1	min = 0.001 max = 99.999 step = 0.001 unit = μm	
	a2	Same as a1	
	m	slct = A or S	
TDRIFT	a1	min = 0.01 max = 99.99 step = 0.01 unit = $^{\circ}\text{C}$	
	a2	Same as a1	
	m	slct = A or S	

Command	Parameter	Limits	Remarks (JBX-9300FS)
PATH	n	min = 1 max = 32 step = 1	
	m	Up to 6 uppercase letters, numbers, and symbols.	Character strings that are reserved as command names cannot be used.
P	i	min = 1 max = 99 step = 1	
	p	A character string can consist of up to 24 uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
	m	slct = MULTI, GATHER, or OVERLAP	
	n	min = 2 max = 10 step = 1	If m is MULTI, max = 5. If m is OVERLAP, max = 8.
SPPRM	m1	min = 0.002 max = 8.000 step = 0.002 unit = μm	Where the accelerating voltage is 50 kV.*
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	min = 0.500000 max = 1.200000 step = 0.000001	
	ss	min = 1 max = 5 step = 1	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	**

* If the acceleration voltage is 25 kV, min = 0.004, max = 16.000, and step = 0.004.

If the voltage is 100 kV, min = 0.00, max = 4.000, and step = 0.001.

** The character size is 1.0 mm $\times$ 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-9300FS)
RESIST	s1	min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	s2	Inapplicable	
	a	Inapplicable	
SHOT	s	min = 1 max = 126 step = even number or = 1	
	t1	min = 40 max = 4000000 step = 1 unit = ns	*
	t2	Inapplicable	
	r	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
MODULAT	t	Up to 6 uppercase letters, numbers, and symbols.	Character strings that are reserved as command names cannot be used.
	r	min = 0 max = 1023 step = 1	
	v	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r0	min = 0 max = r2 step = 1	
	r1	min = 0 max = 1023 step = 1	
	r2	min = r0 max = 1023 step = 1	
	v0	min = -99.9 max = v2 step = 0.1 unit = %	
	v1	min = -50.0 max = 100.0 step = 0.1 unit = %	

* The minimum value depends on the scanning clock of the e-beam lithography machine. For details, refer to the hardware manual of your system. The minimum value mentioned here is a case where scanning clock is 25 MHz.

Command	Parameter	Limits	Remarks (JBX-9300FS)
	v2	min = v0 max = 3200.0 step = 0.1 unit = %	
MMDOFS	i	min = 1 max = 99 step = 1	
	x	min = -25000.000 max = 25000.000 step = 0.001 unit = μm	
	y	Same as x	
	px	Same as x	
	py	Same as x	
	qx	Same as x	
	qy	Same as x	
	rx	Same as x	
	ry	Same as x	
	sx	Same as x	
	sy	Same as x	
	n	min = 1 max = 3168 step = 1	
	cx	min = -25000.000 max = 25000.000 step = 0.001 unit = μm	
	cy	Same as cx	
DEFOCUS	d	min = -4095 max = 4095 step = 1	Inapplicable at present. Designation is ignored.
@	f	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
END	n	min = 1 max = 26 step = 1	*

* The maximum value depends on the parameter m of the MTRL command. If m = STKR, max = 4; if m = INL, max = 1; if m = ALD, max = 2.

5.2.4 For JBX-3030MV

Terminology	
min:	minimum value
max:	maximum value
step	size
unit	unit
slct	selections
fix	fixed value

Table 11 Limits of parameters for JBX-3030MV

Command	Parameter	Limits	Remarks (JBX-3030MV)
MAGAZIN	m	Up to 64 uppercase letters, numbers, and symbols; must begin with a letter.	You cannot specify a character string that includes a single quote (').
MTRL	m	Inapplicable	
#	n	min = 1 max = 6 step = 1	*
	m	Same as n	Same as n
%	s	slct = 2, 3, 4,5 or 6	
	p	An uppercase letter	A to J (10 kinds)
LYR	l	A character string can consists of uppercase and lowercase letters, numbers, and symbols. It can contain up to 64 characters, and must be acceptable as a filename in the computer system.	The character string can contain up to 256 characters in total, if it includes a directory name (or you can specify the character string using a relative path name). If the character string includes a directory name, the directory must contain the file. You cannot specify a character string that includes a single quote (').
JDF	j	Same as LYR l	Same as LYR l
	n	min = 1 max =100 step = 1	
EOS	m	Inapplicable	
	t	Inapplicable	
	c	Inapplicable	
ACC	v	Inapplicable	

* The selectable device depends on the type of the attached substrate handling system. When a mask database for the management of the substrate is used, n is invalidated.

Command	Parameter	Limits	Remarks (JBX-3030MV)
CALPRM	c	A character string of up to 63 characters consisting of printable uppercase and lowercase letters, numbers, and symbols that exclude not only any non-pictorial character including spaces but also slash, backward slash, and single quotation.	
DEFMODE	m	Inapplicable	
RESTOR	m	Inapplicable	
	sw	Inapplicable	
	a	Inapplicable	
OFFSET	dx	min = -140000.000000 max = 120000.000000 step = 0.000001 unit = μm	
	dy	min = -120000.000000 max = 120000.000000 step = 0.000001 unit = μm	
MCOR	gx	min = -100.000000 max = 100.000000 step = 0.000001 unit = $\mu\text{m}/100\text{ mm}$	
	gy	Same as gx	
	ry	Same as gx	
HEIGHT	f	min = -4095 max = 4095 step = 1	
	gx	Same as f	
	gy	Same as f	
	rx	Same as f	
	ry	Same as f	
	sx	Inapplicable	
	sy	Inapplicable	
SUBHEI	gx	Inapplicable	
	gy	Inapplicable	
	rx	Inapplicable	
	ry	Inapplicable	
	shx	Inapplicable	
	shy	Inapplicable	
SHTCOR	l	Inapplicable	
	a	Inapplicable	

Command	Parameter	Limits	Remarks (JBX-3030MV)
	r	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	tx	Same as r	
	ty	Same as r	
STEPPER	s	Inapplicable	
MMDPRM	m	Inapplicable	
SHTOFS	st	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
SPCPRM	spc	Same as SHTOFS st	
PECPRM	pec	Same as SHTOFS st	Same as SHTOFS st
FOGPRM	fog	Same as SHTOFS st	Same as SHTOFS st
PRCPRM	prc	Same as SHTOFS st	Same as SHTOFS st
PRJPRM	prj	Same as SHTOFS st	Same as SHTOFS st
LECPRM	lec	Same as SHTOFS st	Same as SHTOFS st
GLECPRM	gl	Same as SHTOFS st	Same as SHTOFS st
ANG2SIZ	a2s	Same as SHTOFS st	Same as SHTOFS st
SSCPRM	ssc	Same as SHTOFS st	Same as SHTOFS st
PEDPRM	ped	Same as SHTOFS st	Same as SHTOFS st
WARMUP	t	min = 0.1 max = 3276.7 step = 0.1 unit = minutes	
	m	min = 0 max = 2 step = 1	
LBC	sw	Inapplicable	
HSWITCH	swg	Inapplicable	
	swc	Inapplicable	
GLMPRT	ps	min = 0.100 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	fw	min = 0.000000 max = 750.000000 step = 0.000001 unit = μm	
	m	slct = 0 or 1	

Command	Parameter	Limits	Remarks (JBX-3030MV)
PLMPRT	ps	min = 0.100 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	fw	min = 0.000000 max = 750.000000 step = 0.000001 unit = μm	
	m	slct = 0 or 1	
CHMPRT	ps	min = 0.100 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	fw	min = 0.000000 max = 750.000000 step = 0.000001 unit = μm	
	m	slct = 0 or 1	
GLMDET	sw	slct = A, S, M, or C	
PLMDET	sw	fix = A	
CHMDET	sw	fix = A	
	e	slct = 0 or 1	
	n	min = 1 max = 9 step = 1	
WFR3M	m	Inapplicable	
CHIPAL	m	slct = 0, 1, 4, S, V1, or V4	
CDRIFT	a1	min = 0.1 max = 99.9 step = 0.1 unit = %	
	a2	Same as a1	
	m	slct = A or S	
PDRIFT	a1	min = 0.001 max = 99.999 step = 0.001 unit = μm	
	a2	Same as a1	
	m	slct = A or S	
TDRIFT	a1	min = 0.01 max = 99.99 step = 0.01 unit = $^{\circ}\text{C}$	
	a2	Same as a1	
	m	slct = A or S	

Command	Parameter	Limits	Remarks (JBX-3030MV)
PATH	n	min = 1 max = 2000 step = 1	
	m	Up to 64 letters (uppercase and lowercase letters), numbers, and symbols (period "." and underscore "_").	If the character string begins with a character other than a letter, enclose the entire character string in single quotes ('). The character string used for a command cannot be used for another command.
P	i	min = 1 max = 2000 step = 1	
	p	A character string can consists of uppercase and lowercase letters, numbers, and symbols. It can contain up to 64 characters, and must be acceptable as a filename in the computer system.	The character string can contain up to 256 characters in total, if it includes a directory name (or you can specify the character string using a relative path name). If the character string includes a directory name, the directory must contain the file (do not use dtab). You cannot specify a character string that includes a single quote (').
	m	slct = MULTI, MULTI/FP, GATHER, or OVERLAP	You can specify MULTI/FP only in the JBX-3050MV and later models.
	n	min = 2 max = 10 step = 1	If m = MULTI, max = 5 (as maximum). If m = OVERLAP, max = 8 (as maximum)
SPPRM	m1	min = 0.001 max = 2.000 step = 0.001 unit = μm	*
	m2	min = 0.000001 max = 4.000000 step = 0.000001 unit = μm^2	**
	w	min = 0.001 max = 2.000 step = 0.001 unit = μm	*** The values should be multiples of p (1 to 15 times).
	p	min = 0.001 max = 0.250 step = 0.001 unit = μm	

* For the JBX-3050MV, max is limited to 1.000 μm .

** For the JBX-3050MV, max is limited to 1.000000 μm^2 .

*** For the JBX-3050MV, max is limited to 1.000 μm .

Command	Parameter	Limits	Remarks (JBX-3030MV)
	sc	min = 0.500000 max = 1.200000 step = 0.000001	
	ss	min = 1 max = 5 step = 1	
	sfs	min = 1 max = 5 step = 1	
STGPRM	a1	slct = XRASTL, XRASTR, XRASBL , XRASBR, XZIGTL, XZIGTR , XZIGBL, XZIGBR	*
	a2	Same as a1	
	a3	Same as a1	
	a4	Same as a1	
	a5	Same as a1	
C	i	min = 1 max = 4 step = 1	
	s	Up to 64 uppercase letters, numbers, and symbols that can be written.	**
RESIST	s1	min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	s2	Inapplicable	
SHOT	s	Inapplicable	
	t1	Inapplicable	
	t2	Inapplicable	
	r	min = 10.000 max = 150000.000 step = 0.040 unit = ns	
	x	Same as r	
	y	Same as r	
MODULAT	t	Up to 64 letters (uppercase and lowercase letters), numbers, and symbols (period “.” and underscore “_”).	If the character string begins with a character other than a letter, enclose the entire character string in single quotes ('). The character string used for a command cannot be used for another command.

* The STGPRM command is available only in the JBX-3050MV and later models.

** The character size is 1.0 mm × 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-3030MV)
	r	min = 0 max = 63 step = 1	
	v	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r0	min = 0 max = r2 step = 1	
	r1	min = 0 max = 63 step = 1	
	r2	min = r0 max = 63 step = 1	
	v0	min = -99.9 max = v2 step = 0.1 unit = %	
	v1	min = -50.0 max = 100.0 step = 0.1 unit = %	
	v2	min = v0 max = 3200.0 step = 0.1 unit = %	
MMDOFS	i	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
	px	Inapplicable	
	py	Inapplicable	
	qx	Inapplicable	
	qy	Inapplicable	
	rx	Inapplicable	
	ry	Inapplicable	
	sx	Inapplicable	
	sy	Inapplicable	
	n	Inapplicable	
	cx	Inapplicable	
	cy	Inapplicable	
DEFOCUS	d	min = -4095 max = 4095 step = 1	Inapplicable at present. Designation is neglected.

Command	Parameter	Limits	Remarks (JBX-3030MV)
@	f	A character string can consists of uppercase and lowercase letters, numbers, and symbols. It can contain up to 64 characters, and must be acceptable as a filename in the computer system.	The character string can contain up to 256 characters in total, if it includes a directory name (or you can specify the character string using a relative path name). If the character string includes a directory name, the directory must contain the file . You cannot specify a character string that includes a single quote (').
END	n	min = 1 max = 6 step = 1	*

* The selectable device depends on the type of the attached substrate handling system. When a mask database for the management of the substrate is used, n is invalidated.

5.2.5 For JBX-6000FS/E

Terminology	
min:	minimum value
max:	maximum value
step	size
unit	unit
slct	selections
fix	fixed value

Table 5.6 Limits of parameters for JBX-6000FS/E

Command	Parameter	Limits	Remarks (6000FS/E)
MAGAZIN	m	Up to 9 uppercase letters, numbers, and symbols; must start with a letter.	You cannot specify a character string that includes a single quote (').
MTRL	m	Inapplicable	
#	n	min = 1 max = 12 step = 1	*
	m	Same as n	Same as n
%	s	slct = 2, 3, 4, 5, or 6	
	p	An uppercase letter	A to J (10 kinds)
LYR	l	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
JDF	j	Same as LYR l	Same as LYR l
	n	min = 1 max = 99 step = 1	
EOS	m	min = 1 max = 8 step = 1	
	t	Inapplicable	
	c	A character string of up to 63 characters consisting of printable uppercase and lowercase letters, numbers, and symbols that exclude not only any non-pictorial character including spaces but also slash, backward slash, and single quotation.	
ACC	v	Inapplicable	
CALPRM	c	Inapplicable	

* The limit values depend on the type of the attached autoloader. The values mentioned here are those for JBX-6000FS/E.

Command	Parameter	Limits	Remarks (6000FS/E)
DEFMODE	m	Inapplicable	
RESTOR	m	slct = 0 or 1	
	sw	Inapplicable	
	a	Inapplicable	
OFFSET	dx	min = -104900.000 max = 76200.000 step = 0.001 unit = μm	*
	dy	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	Same as dx
MCOR	gx	min = -100.000 max = 100.000 step = 0.001 unit = $\mu\text{m}/100\text{ mm}$	
	gy	Same as gx	
	ry	Same as gx	
HEIGHT	f	min = -4095 max = 4095 step = 1	
	gx	Same as f	
	gy	Same as f	
	rx	Same as f	
	ry	Same as f	
	sx	Inapplicable	
	sy	Inapplicable	
SUBHEI	gx	Inapplicable	
	gy	Inapplicable	
	rx	Inapplicable	
	ry	Inapplicable	
	shx	Inapplicable	
	shy	Inapplicable	
SHTCOR	l	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	a	Same as l	
	r	Inapplicable	
	tx	Inapplicable	

* The limit values depend on the workstage size of the e-beam lithography machine. The values mentioned here are those for JBX-6000FS/E.

Command	Parameter	Limits	Remarks (6000FS/E)
	ty	Inapplicable	
STEPPER	s	Inapplicable	
MMDPRM	m	Inapplicable	
SHTOFS	st	Inapplicable	
SPCPRM	spc	Inapplicable	
PECPRM	pec	Inapplicable	
FOGPRM	fog	Inapplicable	
PRCPRM	prc	Inapplicable	
PRJPRM	prj	Inapplicable	
LECPRM	lec	Inapplicable	
GLECPRM	gl	Inapplicable	
ANG2SIZ	a2s	Inapplicable	
SSCPRM	ssc	Inapplicable	
PEDPRM	ped	Inapplicable	
WARMUP	t	min = 0.1 max = 3276.7 step = 0.1 unit = minutes	
	m	min = 0 max = 2 step = 1	Even if you set m = 0 or 2, m is automatically set to 1.
LBC	sw	slct = ON or HOLD	
HSWITCH	swg	Inapplicable	
	swc	Inapplicable	
GLMPRT	ps	min = 0.010 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	fw	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	m	slct = 0 or 1	
PLMPRT	ps	Inapplicable	
	fw	Inapplicable	
	m	Inapplicable	
CHMPRT	ps	min = 0.010 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	

Command	Parameter	Limits	Remarks (6000FS/E)
	fw	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	m	slct = 0 or 1	
GLMDET	sw	slct = A, S, M, or C	
PLMDET	sw	Inapplicable	
CHMDET	sw	fix = A	
	e	fix = 0 or 1	
	n	min = 1 max = 9 step = 1	
WFR3M	m	min = 1 max = 9 step = 1	
CHIPAL	m	Inapplicable	
CDRIFT	a1	min = 0.1 max = 99.9 step = 0.1 unit = %	
	a2	Same as a1	
	m	slct = A or S	
PDRIFT	a1	min = 0.001 max = 99.999 step = 0.001 unit = μm	
	a2	Same as a1	
	m	slct = A or S	
TDRIFT	a1	min = 0.01 max = 99.99 step = 0.01 unit = $^{\circ}\text{C}$	
	a2	Same as a1	
	m	slct = A or S	
PATH	n	min = 1 max = 32 step = 1	
	m	Up to 6 uppercase letters, numbers, and symbols.	Character strings that are reserved as command names cannot be used.
P	i	min = 1 max = 99 step = 1	

Command	Parameter	Limits	Remarks (6000FS/E)
	p	A character string can consist of up to 24 uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	Single quote (') cannot be included in an ID name.
	m	Inapplicable	
	n	Inapplicable	
SPPRM	m1	Inapplicable	
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	Inapplicable	
	ss	Inapplicable	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	*
RESIST	s1	min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	s2	min = 0.001 max = 1000.000 step = 0.001 unit = nC/cm	
SHOT	s	min = 1 max = 8 step = 1	**
	t1	min = 84 max = 4000000 step = 1 unit = ns	***
	t2	Same as t1	
	r	Inapplicable	

* The character size is 1.6 mm × 1.6 mm, and the writing pitch is 1.6 mm.

** At 25 kV, max = 8 and at 50 kV, max = 16.

*** At JBX-5000LS, min = 167, and at JBX-6000FS/E, min = 84.

Command	Parameter	Limits	Remarks (6000FS/E)
MODULAT	x	Inapplicable	
	y	Inapplicable	
	t	Up to 6 uppercase letters, numbers, and symbols.	Character strings that are reserved as command names cannot be used.
	r	min = 0 max = 63 step = 1	In single line writing, figures greater than 16 are ignored.
	v	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r0	min = 0 max = r2 step = 1	In single line writing, figures greater than 16 are ignored.
	r1	min = 0 max = 63 step = 1	In single line writing, figures greater than 16 are ignored.
	r2	min = r0 max = 63 step = 1	In single line writing, figures greater than 16 are ignored.
	v0	min = -99.9 max = v2 step = 0.1 unit = %	
	v1	min = -50.0 max = 100.0 step = 0.1 unit = %	
	v2	min = v0 max = 3200.0 step = 0.1 unit = %	
MMDOFS	i	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
	px	Inapplicable	
	py	Inapplicable	
	qx	Inapplicable	
	qy	Inapplicable	
	rx	Inapplicable	
	ry	Inapplicable	
	sx	Inapplicable	
	sy	Inapplicable	

Command	Parameter	Limits	Remarks (6000FS/E)
	n	Inapplicable	
	cx	Inapplicable	
	cy	Inapplicable	
DEFOCUS	d	min = -4095 max = 4095 step = 1	Inapplicable at present; designation is ignored.
@	f	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
END	n	min = 1 max = 12 step = 1	*

* The limit values depend on the workstage size of the e-beam lithography machine. The values mentioned here are those for JBX-6000FS/E

5.2.6 For JBX-6300 series

Terminology	
min:	minimum value
max:	maximum value
step	size
unit	unit
slct	selections
fix	fixed value

Table 5.6 Limits of parameters for JBX-6300 series

Command	Parameter	Limits	Remarks (JBX-6300 series)
MAGAZIN	m	Up to 9 uppercase letters, numbers, and symbols; must start with a letter.	You cannot specify a character string that includes a single quote (').
MTRL	m	Inapplicable	
#	n	min = 1 max = 12 step = 1	*
	m	Same as n	Same as n
%	s	slct = 2, 3, 4, 5, or 6	
	p	An uppercase letter	A to J (10 kinds)
LYR	l	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
JDF	j	Same as LYR l	Same as LYR l
	n	min = 1 max = 99 step = 1	
EOS	m	min = 1 max = 8 step = 1	
	t	Inapplicable	
	c	A character string of up to 63 characters consisting of printable uppercase and lowercase letters, numbers, and symbols that exclude not only any non-pictorial character including spaces but also slash, backward slash, and single quotation.	
ACC	v	Inapplicable	
CALPRM	c	Inapplicable	

* The limit values depend on the type of the attached autoloader. The values mentioned here are those for JBX-6300 series.

Command	Parameter	Limits	Remarks (JBX-6300 series)
DEFMODE	m	Inapplicable	
RESTOR	m	slct = 0 or 1	
	sw	Inapplicable	
	a	Inapplicable	
OFFSET	dx	min = -104900.000 max = 76200.000 step = 0.001 unit = μm	
	dy	min = -76200.000 max = 76200.000 step = 0.001 unit = μm	
MCOR	gx	min = -100.000 max = 100.000 step = 0.001 unit = $\mu\text{m}/100\text{ mm}$	
	gy	Same as gx	
	ry	Same as gx	
HEIGHT	f	min = -65535 max = 65535 step = 1	
	gx	min = -131071 max = 131071 step = 1	
	gy	Same as gx	
	rx	Same as gx	
	ry	Same as gx	
	sx	min = -1000000.0 max = 1000000.0 step = 0.1 unit = $\mu\text{m}/100\text{ mm}$	
	sy	Same as sx	
SUBHEI	gx	min = -131071 max = 131071 step = 1	
	gy	Same as gx	
	rx	min = -2047 max = 2047 step = 1	
	ry	Same as rx	
	shx	min = -4095 max = 4095 step = 1	
	shy	Same as shx	

Command	Parameter	Limits	Remarks (JBX-6300 series)
SHTCOR	l	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	a	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r	Inapplicable	
	tx	Inapplicable	
	ty	Inapplicable	
STEPPER	s	Inapplicable	
MMDPRM	m	Inapplicable	
SHTOFS	st	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
SPCPRM	spc	Inapplicable	
PECPRM	pec	Inapplicable	
FOGPRM	fog	Inapplicable	
PRCPRM	prc	Inapplicable	
PRJPRM	prj	Inapplicable	
LECPRM	lec	Inapplicable	
GLECPRM	gl	Inapplicable	
ANG2SIZ	a2s	Inapplicable	
SSCPRM	ssc	Inapplicable	
PEDPRM	ped	Inapplicable	
WARMUP	t	min = 0.1 max = 3276.7 step = 0.1 unit = minutes	
	m	slct = 0,1,2	
LBC	sw	slct = ON or HOLD	
HSWITCH	swg	slct = ON or OFF	
	swc	Same as swg	
GLMPRT	ps	min = 0.010 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	fw	min = 0.000 max = 800.000 step = 0.001 unit = μm	

Command	Parameter	Limits	Remarks (JBX-6300 series)
	m	slct = 0 or 1	
PLMPRT	ps	Inapplicable	
	fw	Inapplicable	
	m	Inapplicable	
CHMPRT	ps	min = 0.010 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	fw	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	m	slct = 0 or 1	
GLMDET	sw	slct = A, S, M, or C	
PLMDET	sw	Inapplicable	
CHMDET	sw	fix = A	
	e	fix = 0 or 1	
	n	min = 1 max = 999 step = 1	
WFR3M	m	min = 1 max = 9 step = 1	
CHIPAL	m	slct = 0,1,4,S,V1,V4	
CDRIFT	a1	min = 0.1 max = 99.9 step = 0.1 unit = %	
	a2	Same as a1	
	m	slct = A or S	
PDRIFT	a1	min = 0.001 max = 99.999 step = 0.001 unit = μm	
	a2	Same as a1	
	m	slct = A or S	
TDRIFT	a1	min = 0.01 max = 99.99 step = 0.01 unit = $^{\circ}\text{C}$	
	a2	Same as a1	
	m	slct = A or S	

Command	Parameter	Limits	Remarks (JBX-6300 series)
PATH	n	min = 1 max = 32 step = 1	
	m	Up to 6 uppercase letters, numbers, and symbols.	Character strings that are reserved as command names cannot be used.
P	i	min = 1 max = 99 step = 1	
	p	A character string can consist of up to 24 uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	Single quote (') cannot be included in an ID name.
	m	slct = MULTI, GATHER, OVERLAP	
	n	min = 2 max = 10 step = 1	If m = MULTI, max = 5 (as maximum). If m = OVERLAP, max = 8 (as maximum)
SPPRM	m1	min = 0.4 max = 4.000 step = 0.002 unit = μm	For EOS mode 2.*
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	min = 0.500000 max = 1.200000 step = 0.000001	
	ss	min = 1 max = 5 step = 1	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	

* The limits depend on the EOS mode. This table shows the limits in each mode.

EOS mode	min	max	step
1 (25 kV-4th lens mode)	0.8	8.000	0.004
2 (50 kV-4th lens mode)	0.4	4.000	0.002
3 (100 kV-4th lens mode)	0.2	2.000	0.001
4 (25 kV-5th lens mode)	0.1	1.000	0.001
5 (50 kV-5th lens mode)	0.05	0.500	0.001
6 (100 kV-5th lens mode)	0.025	0.250	0.001

Command	Parameter	Limits	Remarks (JBX-6300 series)
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min = 1 max = 4 step = 1	
	s	Up to 50 writable uppercase letters, numbers, and symbols.	**
RESIST	s1	min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	s2	Same as s1	
SHOT	s	min = 1 max = 254 step = even number or = 1	
	t1	min = 84 max = 4000000 step = 1 unit = ns	
	t2	Same as t1	
	r	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
MODULAT	t	Up to 6 uppercase letters, numbers, and symbols.	Character strings that are reserved as command names cannot be used.
	r	min = 0 max = 254 step = 1	In single line writing, figures greater than 16 are ignored.
	v	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r0	min = 0 max = r2 step = 1	In single line writing, figures greater than 16 are ignored.
	r1	min = 0 max = 254 step = 1	In single line writing, figures greater than 16 are ignored.
	r2	min = r0 max = 254 step = 1	In single line writing, figures greater than 16 are ignored.

** The character size is 1.6 mm × 1.6 mm, and the writing pitch is 1.6 mm.

Command	Parameter	Limits	Remarks (JBX-6300 series)
	v0	min = -99.9 max = v2 step = 0.1 unit = %	
	v1	min = -50.0 max = 100.0 step = 0.1 unit = %	
	v2	min = v0 max = 3200.0 step = 0.1 unit = %	
MMDOFS	i	min = 1 max = 99 step = 1	
	x	Inapplicable	
	y	Inapplicable	
	px	Inapplicable	
	py	Inapplicable	
	qx	Inapplicable	
	qy	Inapplicable	
	rx	Inapplicable	
	ry	Inapplicable	
	sx	Inapplicable	
	sy	Inapplicable	
	n	Inapplicable	
	cx	Inapplicable	
	cy	Inapplicable	
DEFOCUS	d	min = -4095 max = 4095 step = 1	Inapplicable at present. Designation is ignored.
@	f	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
END	n	min = 1 max = 10 step = 1	*

* The limit values depend on the workstage size of the e-beam lithography machine. The values mentioned here are those for JBX-6000FS/E

5.2.7 For JBX-5500 series

Terminology	
min:	minimum value
max:	maximum value
step	size
unit	unit
slct	selections
fix	fixed value

Table 5.6 Limits of parameters for JBX-5500 series

Command	Parameter	Limits	Remarks (JBX-5500 series)
MAGAZIN	m	Up to 9 uppercase letters, numbers, and symbols; must start with a letter.	You cannot specify a character string that includes a single quote (').
MTRL	m	Inapplicable	
#	n	Inapplicable	
	m	Inapplicable	
%	s	Inapplicable	
	p	Inapplicable	
LYR	l	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
JDF	j	Same as LYR l	Same as LYR l
	n	min = 1 max = 99 step = 1	
EOS	m	min = 1 max = 4 step = 1	
	t	Inapplicable	
	c	A character string of up to 63 characters consisting of printable uppercase and lowercase letters, numbers, and symbols that exclude not only any non-pictorial character including spaces but also slash, backward slash, and single quotation.	
ACC	v	Inapplicable	
CALPRM	c	Inapplicable	
DEFMODE	m	Inapplicable	
RESTOR	m	slct = 0,1	
	sw	Inapplicable	

Command	Parameter	Limits	Remarks (JBX-5500 series)
	a	Inapplicable	
OFFSET	dx	min = -79500.000 max = 50800.000 step = 0.001 unit = μm	
	dy	min = -50800.000 max = 50800.000 step = 0.001 unit = μm	
MCOR	gx	min = -100.000 max = 100.000 step = 0.001 unit = $\mu\text{m}/100\text{mm}$	
	gy	Same as gx	
	ry	Same as gx	
HEIGHT	f	min = -32767 max = 32767 step = 1	
	gx	Same as f	
	gy	Same as f	
	rx	Same as f	
	ry	Same as f	
	sx	min = -1000000.0 max = 1000000.0 step = 0.1 unit = nm	
	sy	Same as sx	
SUBHEI	gx	min = -32767 max = 32767 step = 1	
	gy	Same as gx	
	rx	min = -2047 max = 2047 step = 1	
	ry	Same as rx	
	shx	Same as rx	
	shy	Same as rx	
SHTCOR	l	min = -99.9 max = 3200.0 step = 0.1 unit = %	

Command	Parameter	Limits	Remarks (JBX-5500 series)
	a	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r	Inapplicable	
	tx	Inapplicable	
	ty	Inapplicable	
STEPPER	s	Inapplicable	
MMDPRM	m	Inapplicable	
SHTOFS	st	Inapplicable	
SPCPRM	spc	Inapplicable	
PECPRM	pec	Inapplicable	
FOGPRM	fog	Inapplicable	
PRCPRM	prc	Inapplicable	
PRJPRM	prj	Inapplicable	
LECPRM	lec	Inapplicable	
GLECPRM	gl	Inapplicable	
ANG2SIZ	a2s	Inapplicable	
SSCPRM	ssc	Inapplicable	
PEDPRM	ped	Inapplicable	
WARMUP	t	min = 0.1 max = 3276.7 step = 0.1 unit = minutes	
	m	slct = 0,1,2	
LBC	sw	slct = ON,HOLD	
HSWITCH	swg	Inapplicable	
	swc	Inapplicable	
GLMPRT	ps	min = 0.010 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	fw	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	m	slct = 0,1	
PLMPRT	ps	Inapplicable	
	fw	Inapplicable	
	m	Inapplicable	

Command	Parameter	Limits	Remarks (JBX-5500 series)
CHMPRT	ps	min = 0.010 max = 100000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	fw	min = 0.000 max = 800.000 step = 0.001 unit = μm	
	m	slct = 0,1	
GLMDET	sw	slct = A, S, M, C	
PLMDET	sw	Inapplicable	
CHMDET	sw	fix = A	
	e	slct = 0,1	
	n	min = 1 max = 999 step = 1	
WFR3M	m	min = 1 max = 9 step = 1	
CHIPAL	m	slct = 0, 1, 4, S	Inapplicable at present. Designation is neglected.
CDRIFT	a1	min = 0.1 max = 99.9 step = 0.1 unit = %	
	a2	Same as a1	
	m	slct = A,S	
PDRIFT	a1	min = 0.001 max = 99.999 step = 0.001 unit = μm	
	a2	Same a1	
	m	slct = A, S	
TDRIFT	a1	min = 0.01 max = 99.99 step = 0.01 unit = μm	
	a2	Same as a1	
	m	slct = A,S	
PATH	n	min = 1 max = 32 step = 1	
	m	Up to 6 uppercase letters, numbers, and symbols.	Character strings that are reserved as command names cannot be used.

Command	Parameter	Limits	Remarks (JBX-5500 series)
P	i	min = 1 max = 99 step = 1	
	p	A character string can consist of up to 24 uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	Single quote (') cannot be included in an ID name.
	m	slct = MULTI, GATHER, OVERLAP	
	n	min = 2 max = 10 step = 1	If m = MULTI, max = 5 (as maximum). If m = OVERLAP, max = 8 (as maximum)
SPPRM	m1	min = 1.000 max = 10.000 step = 0.005 unit = μm	For EOS mode 2. *
	m2	Inapplicable	
	w	Inapplicable	
	p	Inapplicable	
	sc	min = 0.500000 max = 1.200000 step = 0.000001	
	ss	min = 1 max = 5 step = 1	
	sfs	Inapplicable	
STGPRM	a1	Inapplicable	
	a2	Inapplicable	
	a3	Inapplicable	
	a4	Inapplicable	
	a5	Inapplicable	
C	i	min = 1 max = 4 step = 1	

* The limits depend on the EOS mode. This table shows the limits in each mode.

EOS mode	min	max	step
1 (25 kV-4th lens mode)	2.000	20.000	0.010
2 (50 kV-4th lens mode)	1.000	10.000	0.005
3 (25 kV-5th lens mode)	0.200	2.000	0.001
4 (50 kV-5th lens mode)	0.100	1.000	0.001

Command	Parameter	Limits	Remarks (JBX-5500 series)
	s	Up to 50 writable uppercase letters, numbers, and symbols.	**
RESIST	s1	min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$	
	s2	a = A ; Area sensitivity min = 0.010 max = 1000000.000 step = 0.001 unit = $\mu\text{C}/\text{cm}^2$ a = L ; Line sensitivity min = 0.001 max = 1000.000 step = 0.001 unit = nC/cm	
	a	A = $\mu\text{C}/\text{cm}^2$ L = nC/cm	
SHOT	s	min = 1 max = 254 slct = even number or = 1	
	t1	min = 84 max = 4000000 step = 1 unit = ns	
	t2	Same as t1	
	r	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
MODULAT	t	Up to 6 uppercase letters, numbers, and symbols.	Character strings that are reserved as command names cannot be used.
	r	min = 0 max = 255 step = 1	In single line writing, figures greater than 16 are ignored.
	v	min = -99.9 max = 3200.0 step = 0.1 unit = %	
	r0	min = 0 max = r2 step = 1	
	r1	min = 0 max = 255 step = 1	In single line writing, figures greater than 16 are ignored.

** The character size is 1.0 mm × 1.0 mm, and the writing pitch is 1.0 mm.

Command	Parameter	Limits	Remarks (JBX-5500 series)
	r2	min = r0 max = 255 step = 1	In single line writing, figures greater than 16 are ignored.
	v0	min = -99.9 max = v2 step = 0.1 unit = %	
	v1	min = -50.0 max = 100.0 step = 0.1 unit = %	
	v2	min = v0 max = 3200.0 step = 0.1 unit = %	
MMDOFS	i	Inapplicable	
	x	Inapplicable	
	y	Inapplicable	
	px	Inapplicable	
	py	Inapplicable	
	qx	Inapplicable	
	qy	Inapplicable	
	rx	Inapplicable	
	ry	Inapplicable	
	sx	Inapplicable	
	sy	Inapplicable	
	n	Inapplicable	
	cx	Inapplicable	
	cy	Inapplicable	
DEFOCUS	d	min = -4095 max = 4095 step = 1	Inapplicable at present. Designation is ignored.
@	f	A character string can consist of uppercase and lowercase letters, numbers, and symbols. It must be acceptable as a filename in the computer system.	You cannot specify a character string that includes a single quote (').
END	n	Inapplicable	

5.3 Writable Characters

Table 5.7 Writable characters

Character	Code	Character	Code
(SPACE)	040	@	100
!	041	A	101
”	042	B	102
#	043	C	103
		D	104
%	045	E	105
&	046	F	106
		G	107
		H	110
		I	111
*	052	J	112
+	053	K	113
,	054	L	114
–	055	M	115
.	056	N	116
/	057	O	117
0	060	P	120
1	061	Q	121
2	062	R	122
3	063	S	123
4	064	T	124
5	065	U	125
6	066	V	126
7	067	W	127
8	070	X	130
9	071	Y	131
:	072	Z	132
;	073	[	133
<	074	\	134
=	075	]	135
>	076	^	136
?	077	_	137

5.4 Examples of Schedules

① Example 1

```
MAGAZIN      ' MAGAZINE1 '
#1-12
LYR          ' SAMPLE01_01 '
END
```

② Example 2ex.2

```
MAGAZIN      ' MAGAZINE2 '
MTRL         STKR
#1
LYR          ' SAMPLE02_01 '
#2
LYR          ' SAMPLE02_02 '
CALPRM       ' CCURRE-A '
#3
LYR          ' SAMPLE03_01 '
P( 1)        ' device001. v30 ', MULTI, 2
P(11)        ' device011. v30 ', MULTI, 2
SPPRM        2. 0, 4. 0, 0. 24, 0. 12, 0. 75, 2
#4
LYR          ' SAMPLE03_02 '
SHTOFS       ' shot_offset '
STEPPER      ' stepper_001 '
END          5
```

③ Example 3

```
MAGAZIN      ' MAGAZINE3 '
#1
%2A
LYR          ' SAMPLE04_01 '
#1
%2B
LYR          ' SAMPLE04_01 '
EOS          1, 0
RESTOR
OFFSET       (5000. 0, -5000. 0)
MCOR         -100. 0, -100. 0, 100
HEIGHT       19, 64, -3, -1, 0
SHTCOR       LINE=10. 0, AREA=-10. 0
WARMUP       5. 0, 1
LBC          HOLD
GLMPRT       3000. 0, 100. 0
CHMPRT       3000. 0, 100. 0
GLMDET       S
CHMDET       A, 1, 3
WFR3M        1
```

```

CDRIFT      10. 0, 5. 0, S
PDRIFT      0. 2, , S
TDRIFT      0. 02, , S
PATH(1)     START
PATH(2)     EXPO-1
PATH(3)     EXPO-2
PATH(4)     END
P(1)        ' JEOL '
C(1)        ' SCHEDULE SAMPLE FOR JBX-6000FS '
RESIST      300. 0, 0. 30
MDLT01:     MODULAT  ((0, -10), (1, 10), (10, 0))
END

```